

Rule-Rewriting Cellular Automata and the Edge of Chaos

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Paper and supplementary material

This packet contains the paper and its four supplements in one file. The paper is self-contained: it states each definition, theorem and empirical finding in prose where it is used, and the supplements carry the formal statements and proofs, the findings notebook, the additional figures, and one exploratory analysis that is reported but not relied upon. Supplement 4 records a measure withdrawn after its mask and measure proved circular; it is included for completeness and is not a premise of the argument.

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Two Readings of the Edge of Chaos

In 2015, working on a different problem, we built a cellular automaton whose cells carried rules rather than states (Corneli & Maclean, 2015). A programming error connected with one example generated spacetime diagrams reminiscent of the edge of chaos behaviour observed in classical CAs. The question of how that example recovered the distinctive visual behaviour remained open.

Recent computational experiments have located that accidentally discovered operator within a larger landscape. A permutation σ of the eight truth-table positions of an elementary rule defines an operator P_σ that rewrites a rule by negating the bits σ names; there are $8! = 40,320$ such operators, and they admit exact classification rather than sampling. They form the bijective core of the 8^8 possible writings. The original example recreated in Figure 3 is a non-bijective writing outside that core. We therefore separate two objects that share a substrate but not a scope: the finite permutation core studied in Part I, and the wider class of state-dependent update architectures studied in Part II.

The classical literature offers two readings of “the edge of chaos”, and they are different kinds of object. The first is a location in *parameter space*, aligned with an order parameter which can be tuned so that the dynamics sit between order and disorder (Langton, 1990). Our classification lands on it from an unexpected direction—every fixed rule of every operator in the family is balanced, with Langton activity $\lambda = 1/2$, exactly the value the classical picture associates with the transition. Yet searching the Meta Cellular Automaton (MetaCA) for a critical transition does not produce one (Supplement 1, Box 6). The algebra places the fixed points precisely at the classical critical value; the dynamics show a broad, drifting crossover and no critical point of that sort at all.

The second reading is not a boundary one tunes toward but a region one measures the system into. It is dynamical: an ordered system absorbs a perturbation, a chaotic one lets it fill the light cone, and the interesting regime lies between. That reading requires no parameter sweep and no choice of where to draw a region—it requires only a perturbation

and a clock. Flipping a single cell and tracking the divergence against an otherwise identical run, on a scale calibrated against elementary rules whose regime is not in dispute, places the phenotype-reading construction of *Constructive control by temporal schedules* between rules 90 and 54, below rule 110 and far below the chaotic rule 30, while frozen rules return exactly zero (*A Causal Measure*). The live phenotype-to-genotype coupling is what puts it there: cutting that single edge, with seed, tape and construction otherwise identical, roughly halves how far a perturbation reaches.

Pursued far enough, the second reading also tests and rejects a tempting substitute for the classical order parameter. Across seven mechanisms, causal reach initially appears to rise and then fall with sustained genotype diversity, but the descending observations obtain high diversity only through mutation or by holding rules still. A conservative eighth construction prescribes the genotype measure and transports it by local bijective exchanges. It sustains causal amplification across the full 32–256 rule range: exchange rate, not diversity alone, controls reach (*Diversity and conservative transport*).

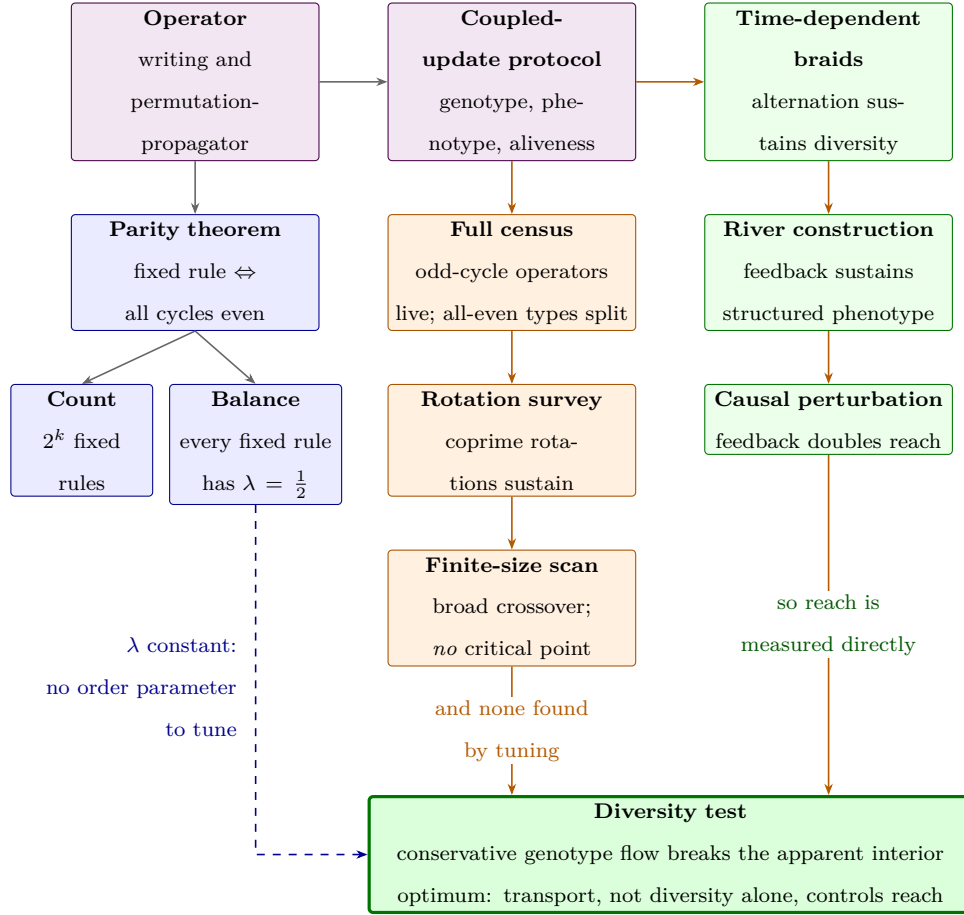
Placing the tested architectures on the calibrated scale then identifies a candidate control within the extension. The constructions separate by *coupling*: where the genotype update never reads the phenotype—the bare operators, braids, niches, refuges, mutation, and blend strength across its whole range—nothing clears the bottom of the complex band, while everything that reads it lands inside (Supplement 1, Box 10). The coupling is moreover a dial rather than a switch. Tuning the gain of that loop carries reach smoothly from the frozen identity past rules 90, 110 and 54, with a locatable crossing; an ungated control matched in mobility, differing only in that its rewriting is blind to the phenotype, never leaves the ordered band. Within these constructions the classical picture is therefore recovered in a coordinate it does not usually use: not a truth-table parameter, but the gain of the loop from the phenotype back to the genotype.

This is the paper’s edge-of-chaos claim, and its scope is the feedback-coupled extension, not the 8! family as a whole. It is measured rather than assumed: in the tested

construction, adding the phenotype-to-genotype edge carries causal reach into the calibrated complex band and cutting it carries reach back.

Supplement 4 reports a separate exploratory analysis of coexisting spatial domains and documents a boundary-transport statistic withdrawn after a circularity check. Neither is used to support either edge-of-chaos claim here.

Part I classifies the permutation-propagators, surveys their spatial dynamics, and tests the parameter-space reading of the edge of chaos. Part II explicitly drops the restriction to a single permutation-propagator and studies causal reach in update architectures built on the same genotype–phenotype substrate. Part III makes the gain of that coupling endogenous rather than imposed, and asks whether a composition whose condition reads the lattice escapes the coordinate Part II identifies. The exact results hold for the entire $8!$ core. The causal measurement covers one principal construction at one lattice width over six seeds, and no comparable causal census of either the $8!$ or 8^8 family is attempted here. Figure 1 maps that division of labour and marks where the scope boundary falls.

**Figure 1**

Argument map and scope boundary. Part I contains the exact branch (blue) and the empirical parameter-space branch (amber) inside the permutation core. The constant value $\lambda = 1/2$ closes off that classical coordinate there; it motivates looking elsewhere but does not entail what follows. Part II is the constructive branch (green): it leaves the single-operator family, introduces time- and phenotype-dependent update architectures, and measures causal propagation. Arrows mark dependence; the change of object between the amber and green branches is explicit.

Part I

The Permutation Core

Definitions, theorems, corollaries and worked examples are stated and proved in Supplement 2, and referred to here by number. The narrative gives each result in prose at the point where it is used, so this part can be read without turning to the supplement; the supplement carries the formal statements, the proofs, and the arithmetic worked by hand.

The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Here each cell also carries a *rule*, and local writes evolve that rule field alongside the phenotype. We display an elementary cellular automaton (ECA) rule in Wolfram order,

$$111, 110, 101, 100, 011, 010, 001, 000,$$

and index these displayed positions from left to right by $p = 0, \dots, 7$. Thus position 0 is the 111 output and most significant bit, while position 7 is the 000 output and least significant bit. Throughout, the indices of σ , w , g , and $\text{bit}[p]$ are these *displayed positions*, not the binary values of the neighbourhoods. If $n \in \{0, \dots, 7\}$ is a numerical neighbourhood value, its rule output is therefore $g(7 - n)$.

We study the operator P_σ , parameterised by a permutation $\sigma \in S_8$ of the displayed positions. Writing the displayed byte as a map $g: \{0, \dots, 7\} \rightarrow \{0, 1\}$, the operator sends g to $P_\sigma g$ defined by

$$(P_\sigma g)(q) = \neg g(\sigma^{-1}(q)), \quad \text{equivalently} \quad \text{bit}[\sigma(p)] := \neg \text{bit}[p]. \quad (1)$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of

$8! = 40,320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified.

The legacy engine stored its table in the order 000, 001, 010, 100, 011, 101, 110, 111. Whenever a legacy writing is imported, the reproduction code conjugates it into the displayed Wolfram order above before we quote its permutation or apply names such as offset +2.¹

Worked example 1 works through one member of the family bit by bit.²

The simultaneous operator acts on one eight-bit rule; the spatial experiments place its elementary writes inside a two-layer automaton.³

The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. That 2015 paper found a single member of this family, encountered through a programming error and not recognised as one of many; we now give the family an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection $\times$ complement, collapsing 256 rules to 88 classes) (Schaller & Svozil, 2025); it is not S_8 , and the negation-carrying, σ -parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν -CA (Dennunzio et al., 2011)) and rules that change during evolution

¹Supplement 2, Definition 1: Writings, elementary writes, and propagators. Supplement 2, Observation 1: Permutations form a small bijective core.

²Supplement 2, Worked example 1: A simultaneous propagator by hand.

³Supplement 2, Definition 2: Genotype and phenotype. Supplement 2, Worked example 2: One site, one complete MetaCA step.

(rule-changing or programmable CA (Mori et al., 1998)). Non-uniform cellular automata are not, by themselves, the source of novelty here: when each site draws its rule from a finite set, the resulting system can be represented as a uniform cellular automaton on a product alphabet (Dennunzio et al., 2011). Our contribution instead concerns the permutation-propagator family and its fixed-point structure.

The Classification

Fixed points exist iff every cycle is even

P_σ has a fixed rule if and only if every cycle of σ has even length; since a one-cycle is odd, this already forces σ to be fixed-point-free.⁴

A fixed rule is a g invariant under (1), i.e. a σ -equivariant Boolean colouring with $g(\sigma k) = \neg g(k)$. Follow a cycle of length L : equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no such colouring, which is why even cycles are required throughout.

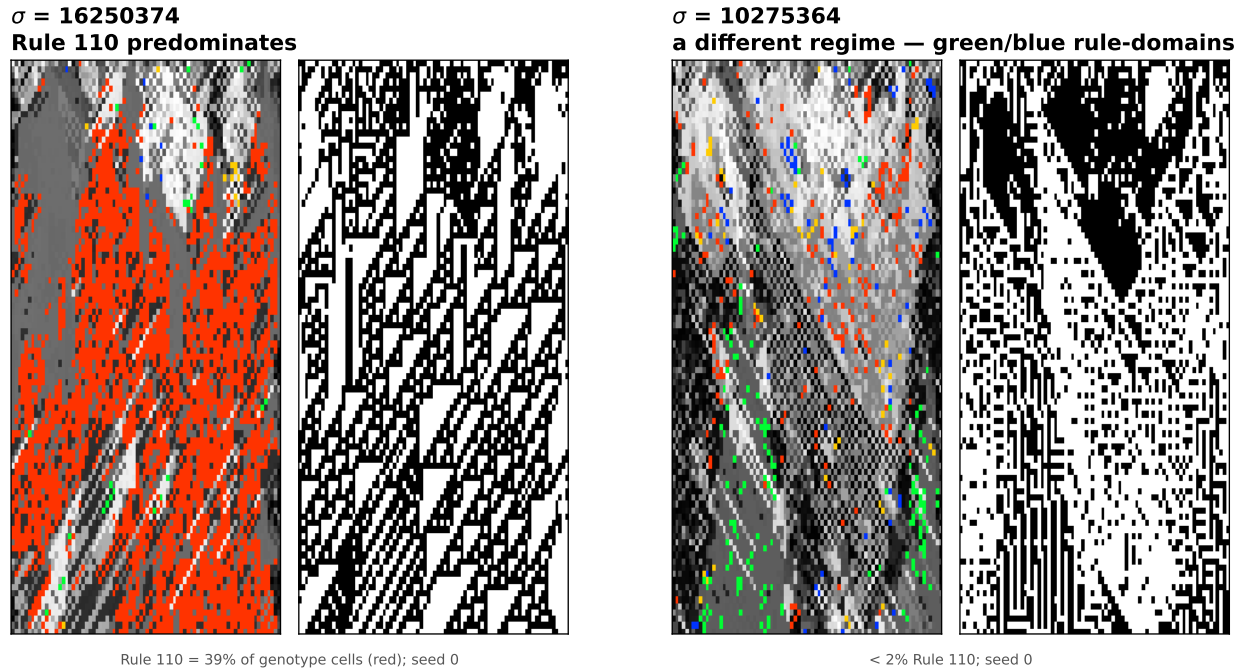
Worked example 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.⁵

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena, 2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two. We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.⁶

⁴Supplement 2, Theorem 1: Fixed points and even cycles.

⁵Supplement 2, Worked example 3: Even and odd cycles.

⁶`DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`; `lake build clean`, no `sorry` or `axiom`.

**Figure 2**

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ the genotype field comes to be dominated by ECA Rule 110—the red cells, 39% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110’s characteristic gliders; under $\sigma = 10275364$ a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine’s own.

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n-1)!!)^2$, with exponential generating function $\sum_n ((2n-1)!!)^2 x^{2n} / (2n)! = 1/\sqrt{1-x^2}$ (the “set of even cycles” construction). For S_8 this is $105^2 = 11,025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

Fixed bytes number 2^k

An all-even σ with k cycles therefore has exactly 2^k fixed bytes, and any σ with an odd cycle has none.⁷ Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k ; the zero case is Theorem 1. This too is machine-checked.⁸ Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40,320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Every fixed point has $\lambda = 1/2$

Every fixed point of every operator in the family has exactly four one-bits—equivalently, Langton’s activity parameter is pinned at $\lambda = 4/8 = 1/2$.⁹ By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$.¹⁰ Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions.¹¹ The statement is machine-checked for Fin 8.¹²

⁷Supplement 2, Corollary 1: The 2^k count.

⁸`alternatingColouring_card_eq_two_pow_cycleType_card`.

⁹Supplement 2, Theorem 2: Every fixed point is balanced.

¹⁰Supplement 2, Definition 3: Langton’s activity λ .

¹¹Supplement 2, Observation 2: Balance is a truth-table property.

¹²`alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ ,

We are deliberate about the standing of this result. Langton’s critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)’s specific genetic-algorithm clustering claim, not the definition of λ , which survived as one axis among others such as Wuensche’s Z (Wuensche, 1999). There is a folklore intuition for why $1/2$ is special — binomial sampling variance $\lambda(1 - \lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule’s dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value $1/2$ is *forced* as an algebraic matter, because any alternation-under-complement colouring is balanced by parity. We claim balance, exactly; any connection of that balance to dynamical behaviour we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ , the fixed points of P_σ —the rules it maps to themselves—number 2^k , where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper’s Figure 8 already exhibited a two-rule residual: a random field of tens of distinct rules dies off to an alternating population of just two, rules 42 and 170. Fittingly, the programming error that produced the family is what pins its residual at two—the erroneous mutation froze a bit, splitting the field into the cells that can reach 42 and those that can reach 170—so the phenomenon was on the page before the 2^k count explained it.

That die-off curve—the count of distinct rules present over time—is a natural

 forcing each to number four.

population-level companion to λ . Where λ is a scalar assigned to a single rule, this curve is a property of the whole field's trajectory; and where λ is uninformative here—every fixed point sits at $1/2$ —the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 6).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Limits of the Tested Distance and Complexity Measures*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule balanced shell at $\lambda = 1/2$.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a single λ ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 4 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R 's four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that holds R among its fixed points (Theorem 1). Whether the field actually settles onto that fixed rule is a separate, dynamical question (as *Fixed Points Are Not Attractors* stresses); here it does. Carried out for Rule 54 this yields

$\sigma = [23015476]$; run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

The parity dichotomy in the spatial dynamics

Theorem 1–2 concern fixed bytes; the spatial dynamics adds stochastic elementary writes and, in the census, neighbour blending. Figure 5 shows why fixed-point existence is not itself a dynamical verdict: without blending, all-even rotate+2 settles within the run while all-even rotate+1 remains diverse; the odd-cycle examples, which lack fixed bytes, also remain diverse. The full census below (*The full census, tabulated by class*) makes this quantitative over all 40,320 operators (Table 1); the dynamical content is the *death-rate difference* between the classes, not the class sizes.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 5, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal

Table 1

Genotype-alive fraction by cycle type under the census protocol, over all 40,320 permutations. Any odd cycle forbids a fixed byte (Theorem 1); empirically, every such operator is alive. The all-even types split, with collapse becoming more frequent as the number of cycles k increases.

cycle type	count	genotype-alive
any type with an odd cycle (17 types)	29,295	100.0%
[8]	5,040	82.9%
[2 6]	3,360	70.0%
[4 4]	1,260	66.7%
[2 2 4]	1,260	49.2%
[2 2 2 2]	105	25.7%

aperiodicity.

The full census, tabulated by class

The trend becomes an exhaustive finite-run observation once every operator is tabulated.¹³ We ran all 40,320 permutation-propagators and grouped them by cycle type (Table 1).¹⁴

Fixed Points Are Not Attractors

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$.¹⁵ Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across $1/2$, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they

¹³Supplement 2, Definition 4: Census protocol and aliveness.

¹⁴Supplement 1, Box 1: The full permutation census.

¹⁵Supplement 2, Worked example 4: Reading the rotation survey.

are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Attraction in the experiments therefore belongs to the stochastic elementary-write dynamics, with or without neighbour coupling, rather than to iteration of the simultaneous map P_σ .¹⁶

Constructive control by temporal schedules

Everything so far has held one operator fixed for the length of a run. Relaxing that—alternating the elementary spatial updates of two operators from step to step, rather than composing them into a single map—turns out to be constructive rather than merely destructive, and it is the first place in the paper where a schedule does something neither of its constituents can.

Braiding two operators that each collapse alone can sustain the field.¹⁷ Figure 8 shows the case: offset +2 (two 4-cycles) and offset +4 (four 2-cycles) are each genotypically dead run on their own, and alive when braided. The schedule is also a usable control rather than a switch. Mixing a sustaining operator with a collapsing one in fixed proportion over a periodic twelve-step schedule moves late-window rule diversity smoothly with the mixing fraction (Figure 9), which is what later lets us treat diversity as a dial to be turned rather than a property a run happens to have.¹⁸

Each operator negates every bit, so under composition the flips cancel in pairs: a product of two is a *pure* coordinate permutation with no negation left.¹⁹

Exploratory searches over products of two and three simultaneous propagators found no additional structural regime.²⁰ Temporal braiding is a different operation: it

¹⁶Supplement 1, Box 2: The rotation survey.

¹⁷Supplement 2, Definition 5: Temporal braid. Supplement 1, Box 3: A braid of two collapsing operators can sustain the field.

¹⁸Supplement 1, Box 4: The mixing fraction is a reproducible diversity control.

¹⁹Supplement 2, Proposition 1: Composition parity.

²⁰Supplement 2, Definition 6: Interrupter.

alternates stochastic elementary writes inside the spatially coupled dynamics rather than replacing them by one composed map.²¹

Algebraic composition is therefore closed in the sense of Proposition 1. Temporal scheduling is different: it retains each intermediate coupled state, which is why the braids of *Constructive control by temporal schedules* can produce dynamics absent from either held operator. A second constructive mechanism pairs a propagator with a step of a different kind.

Limits of the Tested Distance and Complexity Measures

The classification is a conjugacy invariant of σ —its cycle type—while the complexity and compression measures we tried read a single rule’s bits, or one space–time trajectory, without explicitly encoding σ . In our tests none of them recovered the parity classification. A population count reads a collapsed-but-maximally-selected field as nearly dead. A frozen periodic genotype scores a lower statistical complexity C_μ (Crutchfield & Young, 1989) than rule 110. And gzip rates a self-similar rule as noise. We report this as a limitation of the summaries we tested rather than as a finding: it is a negative result about those particular measures, not an impossibility theorem for trajectory-based inference, and unlike the findings collected in Supplement 1 it rests on exploratory runs we did not preserve in reproducible form.

A Parameter-Space Test

The survey (Figure S3.1) exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only the single 8-cycles (offset ± 1 , ± 3) sustain a high-diversity phase, while offset ± 2 and ± 4 collapse. A reader is entitled to ask whether that phase sits at a *critical point*—tuned between an ordered and a disordered phase, as CA complexity is often supposed to (Langton, 1990). This is the first of the two readings distinguished at the outset—an edge of chaos located in *parameter space*, at some value of

²¹Supplement 1, Box 5: The rotation split occurs with and without neighbour blending.

a tunable control parameter—and the finite-size scan below is the test of it, the only place in this paper where that reading is put to the question. We then ask whether the sustained diversity is spatially organised into coexisting dynamical regimes.²²

Figure 10 collects the four diagnostics. The scan is over the propagator fraction q , from all blend at $q = 0$ to all propagator at $q = 1$, at lattice widths $L = 30$ to 240 with 32 seeds each; the point of varying L is that a genuine transition sharpens as the system grows while a crossover does not. Genotype activity fits a logistic in q at every size ($R^2 > 0.99$), but the crossover width does not shrink over these sizes ($w \sim L^{-0.04}$). The size-scaled run-to-run spread $L \text{Var}(a_G)$, used here as a susceptibility proxy, does peak inside the band and grows about 5.7-fold, but its maximum wanders—from $q = 0.75$ to 0.50 and then to the sampled boundary $q = 1$ for the two largest sizes—rather than settling. Finite-horizon collapse becomes rarer with size, and the Binder cumulant has no common crossing. Each diagnostic is individually consistent with a critical point; none of them locates one, and together they support a broad crossover instead.

The genotype field is also spatially organised. Tinting by activity resolves it into coexisting low- and high-activity domains whose boundaries trace fractal-like filamentary walls through spacetime; those walls carry elevated genotype churn, and at matched activity the two sides hold measurably different rule populations, so they are not an artefact of the tint that draws them. We report that structure in Supplement 4, together with a measure of information transport along those boundaries that we built, validated against three nulls, and then withdrew when it proved to be an artefact of its own construction. Neither is a third edge, and nothing in the argument below depends on them.

²²Supplement 1, Box 6: No finite-size evidence of a critical point over the tested range.

Part II

Beyond the Permutation Core: Feedback and Causal Propagation

Part I's classified object and exhaustive census were the finite permutation core; its temporal braid was the boundary case, already replacing one held operator with a schedule. We now retain the same genotype–phenotype lattice and elementary rule semantics but drop the single-operator restriction. The objects in Part II are update architectures: time-dependent schedules, phenotype-conditioned writes, mutation and holding protocols, and conservative local transports. Some use writings from the 8^8 family as components, but a state-dependent architecture is not itself one fixed writing. Consequently, the results below neither classify nor measure the $8!$ family as a whole; they test how causal propagation changes when the genotype update can read the phenotype.

A Feedback-Coupled Construction

The next construction crosses that boundary by allowing the update to depend on the live phenotype. Figure 11 shows a representative run at $L = 240$: alternating phenotype domains branch, merge and reorganise for the length of the run rather than resolving to a homogeneous or short-period field.²³

A Causal Measure

A causal test also avoids a general problem with observational regions. Any statistic computed over a region drawn from the field it is computed on invites circularity, and no null defined only by moving that region can necessarily detect it. We therefore stop drawing regions: perturb one cell and ask where the effect arrives.

We run the phenotype-reading construction of *A Feedback-Coupled Construction* to

²³Supplement 2, Definition 7: River. Supplement 1, Box 7: The river sustains structured phenotype.

$t^* = 60$, fork, flip a single phenotype bit at site x , and continue both branches to $T = 120$, recording which cells differ. The fork is exact: the construction draws one pseudo-random position per cell per step, so re-running from the same seed gives both branches an identical tape and every divergence is the causal consequence of the flip. The control is the matched feedback-off variant—same seed, tape, construction and initial state, with only the live phenotype-to-genotype edge cut. Figure 12 sweeps the flip over all 80 sites and records damage at a sequence of horizons.²⁴

Two readings of that result must be separated. A glider carries a fixed-width packet at constant velocity; a conductive medium disperses. Tracking the damage centroid distinguishes them, and the reference values must be measured rather than assumed—elementary rule 110, whose gliders support universal computation, does not give a fixed packet under a random single-site perturbation, because the perturbation creates and destroys gliders which then collide.²⁵

Figure 13 places every construction on that one axis. All points share the protocol of Supplement 1, Box 9, and the calibration reproduces rules 0, 204 and 90 exactly against it, so the bands drawn from the elementary rules carry over unchanged. The rows split the constructions by coupling rather than by provenance, and the split is clean. Nothing whose genotype update is blind to the phenotype clears the bottom of the complex band—not even the ungated transport controls, which are fully mobile and merely cannot see what they are moving through. Everything that reads the phenotype lands inside. The river appears in both rows, at 12.97 with its edge and 5.51 without, so the arrow between them is one construction moved by adding an edge rather than by changing a parameter.

The same sweep also gives the growth of the damage cone and the displacement of its centroid (Figure 14); we report those exponents descriptively and draw no inference

²⁴Supplement 1, Box 8: Phenotype-to-genotype feedback roughly doubles causal reach.

²⁵Supplement 1, Box 9: Causal reach on a scale calibrated against elementary rules. Supplement 1, Box 10: The order parameter is the strength of the phenotype-to-genotype coupling.

from them about gliders or particles.

The coupling as a dial

Treating that coupling as present or absent understates it. In both constructions that read the phenotype it is a quantity, and causal reach is a graded, monotone function of it.²⁶

For conservative transport the gain is the phenotype-gated swap rate; reach crosses rule 90 between rates 0.10 and 0.20 and saturates past 0.75. For the river it is the *currency* of the phenotype bits the genotype reads: each cell at each step reads the live phenotype with probability γ and a frozen reference otherwise, so γ is the fraction of the genotype's view that is causally current. Because the frozen field is itself a real phenotype, marginals and spatial structure are matched at every γ and only causal currency varies; because $\gamma = 1$ reproduces the live river exactly, the dial is anchored against a measurement made independently of it.

Two consequences bear on the reading of *A Causal Measure*. First, the $\gamma = 0$ endpoint is 1.28, not the 5.51 of the ablated control: that control freezes the phenotype the genotype reads but still presents a field which, in the perturbed branch, contains the flip. It cuts the *currency* of the read, not the read. The doubling reported for it is therefore the smaller part of a coupling effect that spans an order of magnitude.

Second, the two dials have opposite curvature (Figure 15). Transport is concave and buys most of its reach early; the river is convex, with 45% of its span in the last eighth of γ , so one stale read in eight costs 40% of the reach. Monotonicity in the coupling is thus common to both constructions while its shape is not, and we would not generalise the shape from two cases.

²⁶Supplement 1, Box 11: Coupling is a dial, not a switch, and reach is monotone along it.

Diversity and conservative transport

Langton’s λ is silent on this family: the classification puts *every* fixed rule of *every* operator at $\lambda = 1/2$, so it cannot distinguish an operator whose field stays diverse from one that collapses. It cannot be rescued by computing it on the realized dynamics either.

Across the nine configurations of Figure S3.2 the empirical neighbourhood distribution is close to uniform—normalised visitation entropy 0.946 to 0.999, all eight neighbourhoods visited in every run—so λ read off the realized transitions with uniform weight (0.42 to 0.52) and the same quantity weighted by visitation (0.40 to 0.51) differ by 0.035 on average, and both sit near $1/2$ wherever the nominal value does. Nor is the silence peculiar to this family: of the elementary rules calibrating the scale above, rules 30, 54, 90, 150, 204 and 232 all have $\lambda = 1/2$ exactly, spanning frozen, ordered, complex and chaotic. What the balance theorem contributes is not a new failure of λ but an exact and provable instance of a known one. The sustained distinct-rule count can, and the causal measure above lets us ask the stronger question: does causal reach depend on diversity itself, or on the mechanism used to sustain it?

Our initial survey varied sustained diversity by seven mechanisms that share little: the bare operator family; braided writings; the fork time after seeding the genotype with a randomised *full* rule population; a continuous blend strength; random rule mutation; asynchronous refuges that hold a cell’s rule exactly with some probability; and spatially phase-shifted braid niches. A limiting case anchors the top of the range: a genotype held fixed at the full population has maximal diversity and no rewriting at all. The coordinate throughout is diversity *sustained through the response window*, not diversity at the instant of the fork—a field seeded with all 256 rules holds them for one step and collapses during the very window the damage is measured in.²⁷

Figure 16 shows why the apparent optimum was an artefact. The original

²⁷Supplement 1, Box 12: Sustained diversity does not determine causal reach; conservative transport breaks the apparent interior optimum.

configurations trace a rise and fall against sustained diversity, but every point on the descending limb obtains its diversity either by replacement mutation or by holding rules still, so the descent measures those two mechanisms rather than diversity. The conservative runs, which hold the rule histogram exactly invariant while varying transport, separate into flat bands — one per transport rate — that span the whole diversity axis. Reading across a band shows the effect of diversity at fixed transport and finds almost none; reading between bands shows the effect of transport and finds an order of magnitude.

Two cautions bound this result. First, the conservative intervention is a local transposition, rather than the one-bit, one-site intervention of the original survey. Normalising by initial damage puts their amplification on a common scale, but the conservative curves are a falsification control, not a claim that their absolute heights rank all mechanisms. Second, phenotype-gated bijective exchange is one deliberately constructed transport law, tested at one lattice width and one response horizon. It establishes existence: fields can remain mobile and propagate damage at effective diversities well above 135 without mutation. It does not establish that transport rate is a universal order parameter.

The defensible conclusion is narrower and cleaner. Distinct-rule diversity describes population persistence in the Part I operator experiments, but it does not govern causal reach across the Part II architectures. The latter depends on how rewriting or transport is organised. Identifying a coordinate that captures that organisation across mechanisms is work for another paper; the classification of *The Classification* guarantees only that λ cannot discriminate among fixed points in the permutation core.

What this scale does and does not license needs saying plainly. It separates systems that propagate a perturbation from those that do not, and that separation is sharp: frozen rules return exactly zero. It is *not* a regime coordinate. Wolfram class III rules bracket the range at both ends—rule 90 at 8.00 and rule 30 at 37.50—with the class IV rules 54 and 110 between them, so a position on this axis does not by itself classify a system as ordered, complex or chaotic, and we make no such classification. What we claim is narrower and

causal: the live phenotype-to-genotype edge roughly doubles how far a single-cell perturbation reaches, on a measure that involves no region-drawing and therefore cannot fail in the way the withdrawn measure of the Supplement 4 analysis failed. It concerns the phenotype-coupled construction, not the bare operators; no comparable measurement of the family is attempted here.

What the coupling supplies

One measurement bounds how the convexity of *The coupling as a dial* should be read. Asking, per cell and step, whether the live phenotype context selects a *different* rule than a frozen context would: it does in 73.1% of cell-steps (and in 76.8% against no context), over four seeds at $L = 80$ for 120 steps. The construction's first-match runs over four candidates, so an uninformative re-selection would differ in about 75% of cases: at 73.1% a frozen context carries almost no information about the live answer. The read is determinative rather than a tiebreak, which is what a convex gain curve should lead one to expect, and it means there is no common case that a phenotype-blind rule could encode in its place. We draw no stronger conclusion from this than that: whether *any* blind construction could match the river is a question about a search we have not run, and Supplement 1, Box 8 reports only that sixteen hand-chosen ones do not.

Part III

Endogenous Coupling

In Part II the gain of the phenotype-to-genotype loop is a parameter we set. This part asks what happens when it is not: when the coupling is a function of the lattice's own local state rather than a dial turned from outside. The objects are still compositions of operators Part I classifies, so nothing new is introduced at the level of the substrate; what changes is who sets the gain.

An Endogenous Gain

A *conditional composition* applies one operator where a local predicate holds and another where it does not: $\text{switch}(c, A, B)$. Throughout this part A is conservative transport at unit rate, which sits in the chaotic band of the calibration of Section II at 25.38, and B is holding, which sits in the ordered band at 1.21. The predicate decides, cell by cell and step by step, which applies. Its *firing fraction* f —the proportion of opportunities at which A is selected—is measured from the run rather than prescribed.

Two disciplines make the comparison below meaningful rather than merely suggestive, and both are inherited from the gain dial of Section II. First, the transport stream advances once at every opportunity whether or not the gate fires, so a predicate that reads the phenotype cannot desynchronise the two branches of the damage fork and manufacture divergence that is not causal. Second, where the predicate consumes randomness of its own, those draws come from a separate stream re-seeded identically in both branches, so the gate never itself injects difference.

The first question is what a composition does when its condition carries no information about the lattice. A gate firing at a prescribed rate, blind to the phenotype, is well described across six rates spanning $f = 0.10$ to $f = 0.94$ by a line through the holding value,

$$\text{reach}(f) = 1.21 + 22.42 f, \quad (2)$$

with the unconstrained fit returning $R^2 = 0.993$. A blind conditional composition is therefore *exactly* a gain dial: its duty cycle plays the role γ plays in Section II, and reach is linear in it. This is worth stating plainly because it is the null against which everything else in this part is measured, and because it means the construction adds nothing on its own. Making the gain a function of *something* does not by itself buy anything; it matters what that something is.

A Reading Gate Exceeds the Dial

Now let the predicate read the phenotype. Write $\text{agree } k/m$ for the condition that at least k cells of the centred circular m -cell phenotype neighbourhood agree with the centre. Thirteen such gates were placed on the same scale, spanning widths $m \in \{3, 5, 7, 9\}$ and, at each width, strictness from bare majority to unanimity.

Every one of them exceeds the blind line of Equation 2 at its own measured firing fraction, and several do so by a wide margin. The largest departure is +22.8 cells; the widest permissive gate, $\text{agree } 5/9$, reaches 38.23 while firing on only 63% of opportunities—not merely above the blind line but above unit-rate transport’s own 25.38. Firing a chaotic operator less often, in the right places, spreads damage further than firing it always.

What predicts the size of the departure is the *width* of the neighbourhood the gate reads, not how hard the gate is to satisfy. Adding width to a quadratic-in- f model of departure raises R^2 from 0.84 to 0.96, at +1.47 cells per cell of width; adding strictness instead raises it only to 0.88. The design separates the two—the correlation between width and strictness across the thirteen gates is +0.01—whereas strictness is nearly collinear with the firing fraction itself at -0.91 , so its apparent effect is largely f under another name. The model-free version of the same statement is that within each of three groups of gates matched on f , departure increases monotonically with width; in the group near $f \approx 0.65$ the strictness ratio is almost constant at 0.56–0.60 while the width nearly doubles.

Currency, Not Correlation

That result admits a deflationary reading, and it is the reading its own mechanism suggests. A gate reading m cells has neighbouring decisions sharing $m - 1$ inputs, so width *is* the spatial correlation length of the firing pattern. The blind gate of Equation 2 draws an independent coin per cell and is therefore spatially uncorrelated. Correlated firing of a *transport* operator composes adjacent swaps into longer-range movement, so a departure that grows with width might be pure geometry, with the phenotype playing no part beyond

supplying a conveniently correlated field.

The control that separates the two possibilities holds the phenotype still. A *frozen* gate applies the identical agree k/m predicate to the phenotype as it stood at t^* : same predicate, same width, same spatial statistics, a real phenotype field, and therefore the same correlation structure in its firing. What it cannot do is track where the perturbation has since spread. Each branch of the damage fork keeps its own t^* snapshot, so both frozen fields contain the perturbation itself and only the live gate sees its consequences—the ablation of Section II applied to the gate rather than to the update.

Across seven pairs matched on (m, k) , the live gate exceeds its frozen counterpart in every case, by 14.5 cells on average. More tellingly, every frozen departure from the blind line is *negative*, between -0.04 and -4.18 . A gate with the same width and the same correlated firing, reading a field that is merely stale, sits at or slightly below the spatially uncorrelated baseline. Spatial correlation on its own buys nothing here; if anything it costs a little. The comparison is also conservative in the direction that matters: the frozen gate fires *more* often than its live counterpart in six of the seven pairs, so it is given more opportunities and still reaches far less.

The reading we take from this is that the gate is a targeting mechanism, and targeting works only on current information. Firing transport where the phenotype is *now* locally uniform lifts causal reach far above a blind gate at the same rate; firing it where the phenotype *was* uniform at t^* does not, though the two firing patterns share width, spatial statistics and rate. So the coordinate that governed in Part II governs here too, reached by a different route: an endogenous gain does not escape the causal currency of what the genotype reads, it re-derives it. A composition whose condition is blind reproduces the dial exactly; a composition whose condition reads the phenotype exceeds the dial precisely to the extent that what it reads is current.

Discussion

Limits

The Part I theorems are exact for the Boolean permutation core; its dynamical claims are narrower and empirical. The census uses three seeds per operator, the rotation survey two, and the finite-size scan 32 per parameter cell, all under the stated radius-one spatial protocol. These finite-run results do not turn fixed-point balance into a claim about attraction. The substrate itself is not new (*The Object*), and λ is one heuristic among several (Wuensche, 1999); what is exact is only that the fixed-point structure of Equation (1) forces every fixed rule in the core to have $\lambda = 1/2$.

Part II has different limits because it has a different object. Its central causal comparison concerns one feedback-coupled construction rather than either writing family, one lattice width, and six seeds. The descriptive exponents are fitted over a range where damage width reaches only a tenth of the lattice, so finite-size effects are not excluded. The architecture survey and conservative-transport control compare mechanisms rather than exhaustively sampling 8^8 writings, and their fixed sample sizes are stated with each result. They identify feedback and transport as controls in the tested constructions, not as universal order parameters. Illustrative diagrams use pinned representative seeds. All reported computations are reproducible from the accompanying package.

Supplement 4 carries the separate spatial-domain analysis and the information measure withdrawn after its mask and measure proved circular. Its claims and failures do not enter either Part I's classification or Part II's causal comparison.

Conclusion

The paper has studied three related objects, and its conclusions must keep them separate. Part I concerns the finite permutation core. Its static results are exact: a permutation-propagator possesses fixed rules exactly when all cycles are even; an all-even permutation with k cycles has 2^k fixed rules; and every such rule is balanced, with $\lambda = 1/2$.

Its dynamical results are finite-run observations over that core. The exhaustive census shows that all odd-cycle operators remain genotype-alive under the stated protocol while every all-even cycle type contains both sustaining and collapsing members. The rotation survey and finite-size scan sharpen the distinction: fixed-point existence does not determine attraction, and the tested propagator-fraction scan gives a broad crossover rather than evidence of a critical point.

Part II retains the two-layer lattice but changes the object from an individual writing to an update architecture. The river construction composes a permutation-propagator with a phenotype-reading step; the wider comparison adds temporal schedules, mutation, holding, niches, and conservative local transport. These are not a census of the 8^8 writings, and the results do not retroactively characterise the $8!$ core. They establish a narrower causal claim. Against a matched feedback-off control, phenotype-to-genotype feedback roughly doubles the reach of a single-cell perturbation. When the strength of that feedback is varied, reach changes monotonically, while a control matched in rule mobility but blind to the phenotype remains in the low-reach band. Conservative transport further shows that sustained genotype diversity is not itself the governing coordinate: high diversity can coexist with propagation, and transport changes reach far more than diversity does.

The connection between the parts is therefore motivational rather than deductive. Part I proves that the classical λ coordinate cannot discriminate among fixed points in the permutation core and finds no critical point in the tested parameter scan. Part II asks what a causal measurement can reveal once the architecture itself is allowed to change. Its answer is feedback coupling for the constructions tested here, not a theorem about all permutations or all writings.

This admits a reading the substrate makes unusually clean. The gain γ interpolates between a genotype that reads only an ancestral phenotype and one that reads the currently acquired state, with the frozen field a real phenotype so that marginal statistics

and spatial structure are matched at every setting; only causal currency varies. What the dial varies is thus how legible acquired state is to the heritable layer—the Lamarckian question, posed as a continuum rather than a dichotomy. Its discriminating power is unusual in this family: across λ , propagator fraction, sustained rule diversity, mutation rate, refuge probability, niche width and rule mobility, causal reach never leaves the ordered band, and blend strength alone reaches its edge. The substrate carries no fitness, so this concerns what the heritable layer can see and not what is selected for.

Part III changes the object once more, from an architecture whose gain we set to one that computes its own. Its results are narrower than Part I's and of the same kind as Part II's: finite-run causal measurements over a chosen family of predicates, not a characterisation of conditional compositions in general. What they establish is a negative and a positive that belong together. A composition whose condition is blind to the phenotype reproduces the Part II dial exactly, linear in its duty cycle—so conditional composition is not by itself a new mechanism. A composition whose condition reads the phenotype exceeds that dial, by margins that grow with the width of the neighbourhood read; but an otherwise identical gate reading a phenotype frozen at t^* does not exceed it at all, and in every tested pair falls slightly below. Width is the spatial correlation length of the firing pattern, so this is the comparison that matters: correlated firing alone buys nothing, and what the reading gate exploits is that what it reads is current. The coordinate is therefore the same one Part II identifies, recovered from a construction that shares none of Part II's dials. That convergence is the strongest connection the paper draws between its parts, and it is still a convergence of measurements rather than a theorem. Supplement 4 records the separate spatial-domain analysis and the withdrawn boundary statistic; neither is a premise of this two-part argument.

Part III is narrower still. Its measurements cover one pair of constituents—unit-rate transport against holding—and one predicate family, agreement within a centred circular neighbourhood. Thirteen conditioned gates at four widths support the width gradient;

seven matched pairs support the frozen control. Each row is sixteen seeds at one lattice width, so the between-seed spread is real and reported rather than smoothed. Two limits deserve naming. The frozen gate’s firing fraction does not match its live counterpart exactly, differing by up to 0.04; the discrepancy runs in the direction that weakens the result, since the frozen gate mostly fires more often and still reaches less, but it is not zero. And while the frozen control shows that current information is *necessary* for the departure, it does not decompose the width gradient: how much of that gradient is currency and how much is the geometry of correlated firing is not resolved here. A blind gate with matched correlation length, rather than matched rate alone, would separate them and has not been run.

Future work. Part III closes the first of the two extensions the gain result suggested and sharpens the second. The gain is no longer only a parameter we set: a predicate reading the lattice sets it, and reach responds to whether what the predicate reads is current. What that leaves open is the step from a gate to a regulator. The conditions tested here are fixed families chosen by us; nothing yet adjusts its own condition in response to how it is doing. The 2014 ancestor of this substrate already contains the relevant construction—a switch that fires only where the local context carries no new information—so a system that tunes its own gate is buildable on this lattice rather than hypothetical, and the question is whether such a system settles anywhere in particular on the calibrated scale.

The second extension is unchanged and remains the more interesting. As it stands the substrate instantiates the Lamarckian arm alone: acquired state writes into the heritable layer, and both the dial of Part II and the gates of Part III vary the strength or the aim of that write-back. The Baldwin effect requires something this substrate lacks—a selective filter, so that plasticity alters what selection sees and the heritable layer follows without direct transcription. Adding selection over a population of these architectures would supply that ingredient, and because both arms would then be expressible in one

family they could be placed on the single measure used throughout this paper: the reach of damage-spreading, calibrated against rules of undisputed regime. Whether direct write-back and selection-mediated assimilation arrive at the same place on that scale, and by what route, is a question the family is now equipped to ask.

Data and code availability. Every numerical result in this paper is regenerated from source rather than deposited: the generators are seeded, so a single documented command rebuilds the fields underlying the tables, figures, and reported summary statistics from an empty data directory, and the analysis scripts consume no input that command does not produce. That command was run twice from an empty data directory and all 307 generated artefacts were byte-identical across the two runs; the 298 of them that also exist as published artefacts match those as well. The code and those commands are publicly available at <https://github.com/tothedarktowercame/mmca-clj> (commit 7e04a6de6afe35de041b0ac4bb52a5cea25fe8d4)—a standalone, dependency-light Clojure port of the MetaCA dynamics whose `README` gives the complete reproduction command and commands for individual figure groups. The classification and balance theorems are formally verified in Lean 4 with Mathlib at <https://github.com/tothedarktowercame/mathlib4>, branch `darktower` (commit bc34d8cec27216aae1d3cd511a9950ab1bce3575), in `DarkTower/Patterns/Propagator.lean`: the cycle-parity criterion as `hasAlternatingColouring_iff_cycleType_even` and the $\lambda = 1/2$ balance as `alternatingColouring_true_card_eq_four`. The same file also contains executable regressions for both hand calculations, including the displayed-position versus numerical-neighbourhood indexing contrast.

Generative-AI disclosure. Anthropic Claude (Opus 4.8, Opus 5, and Fable 5) and OpenAI Codex (version 5.6-sol) coding agents (Anthropic, 2026a, 2026b; OpenAI, 2026) assisted in an iterative, human-supervised workflow with manuscript editing, software implementation, test construction, and adversarial checks. Exchanges with Claude

Fable 5 contributed to the initial identification of the operator family reported here. The author reviewed the resulting prose and code, reran the reported computations and formal checks, and accepts full responsibility for the manuscript. Product information was accessed in July 2026; per-source access dates are given in the bibliography.

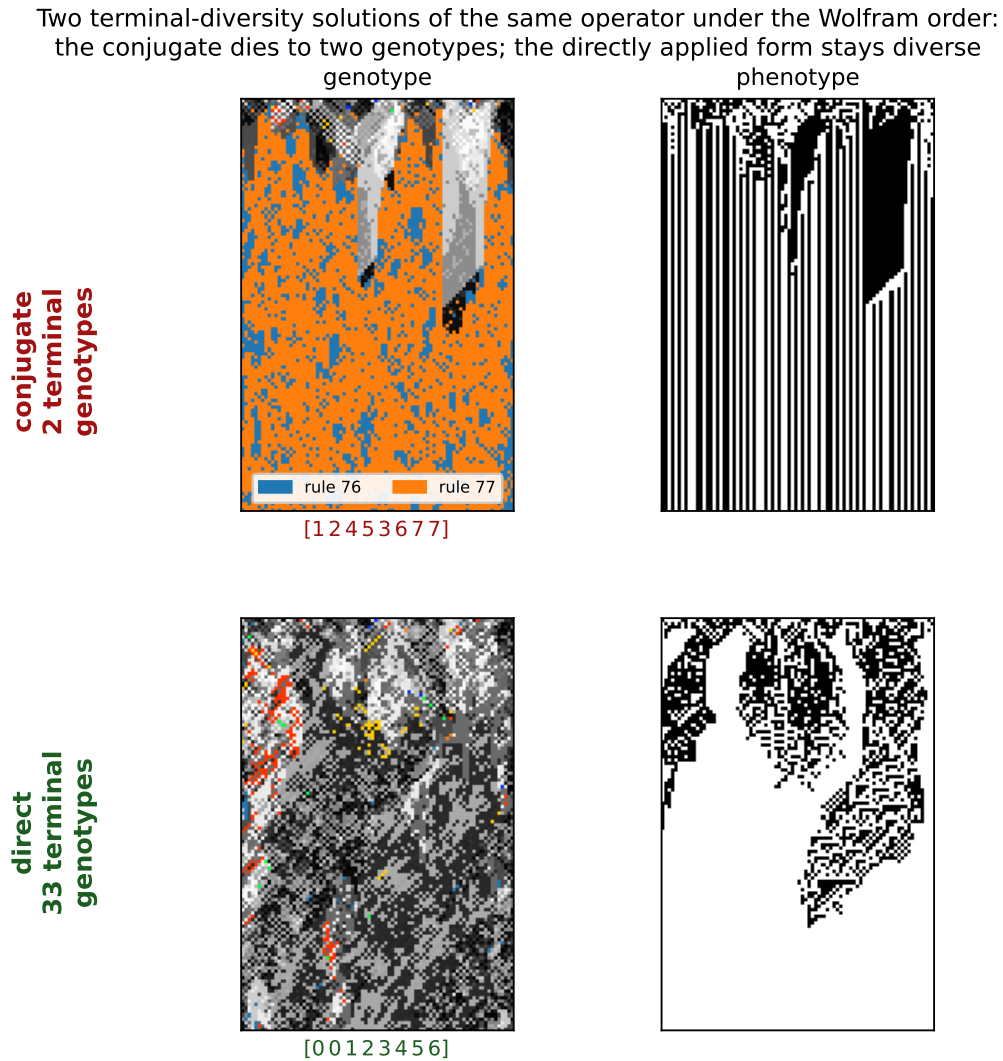
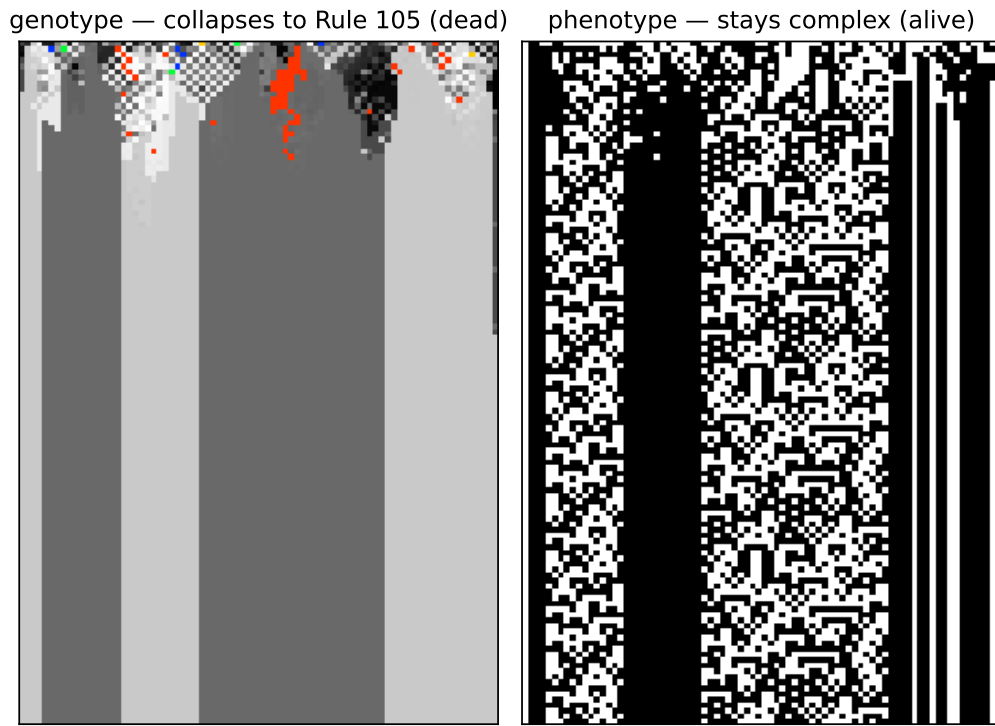


Figure 3

Two terminal-diversity solutions of one historical operator. The historical write [00123456] has two forms under the standard Wolfram order. Its conjugate [12453677] (top) reproduces the two-rule behaviour: the genotype collapses to exactly two terminal rules—76 and 77, adjacent, differing in a single bit—regardless of seed, while the phenotype stays regular. Applied directly (bottom), the same operator keeps a diverse genotype (~30 terminal rules). Two ends of genotypic aliveness from one historical write.

**Figure 4**

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [23015476]$, built to fix the balanced shell containing Rule 105, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule 105, an additive $\lambda = 1/2$ rule whose dynamics keep it complex. Seed 1, $W = 80$, $T = 120$.

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even

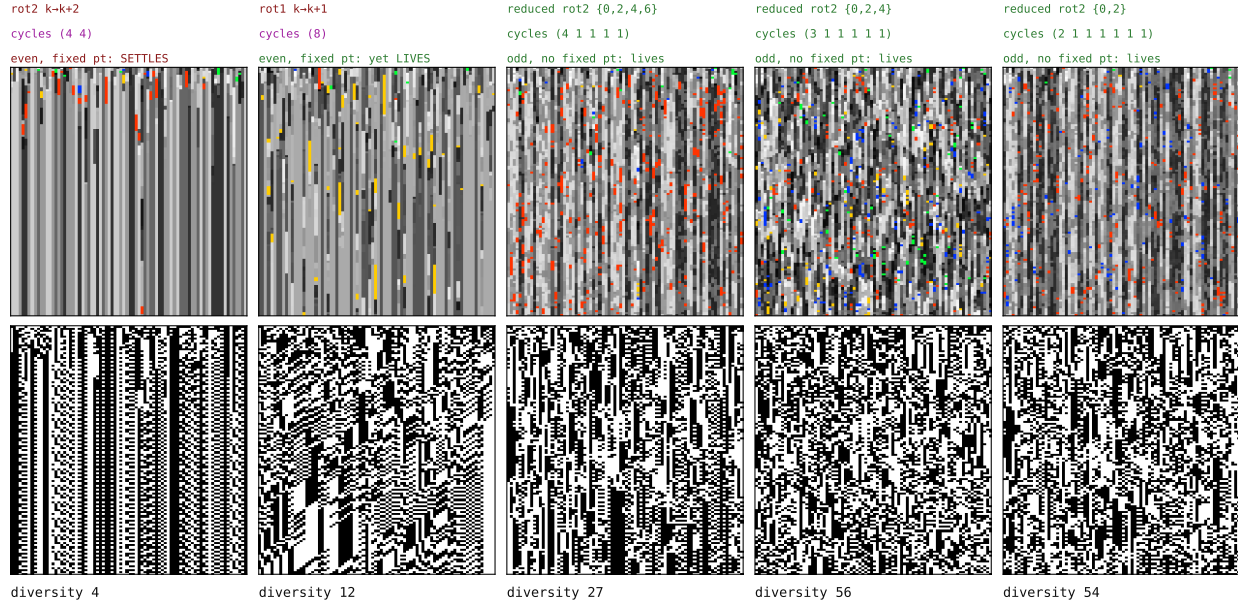
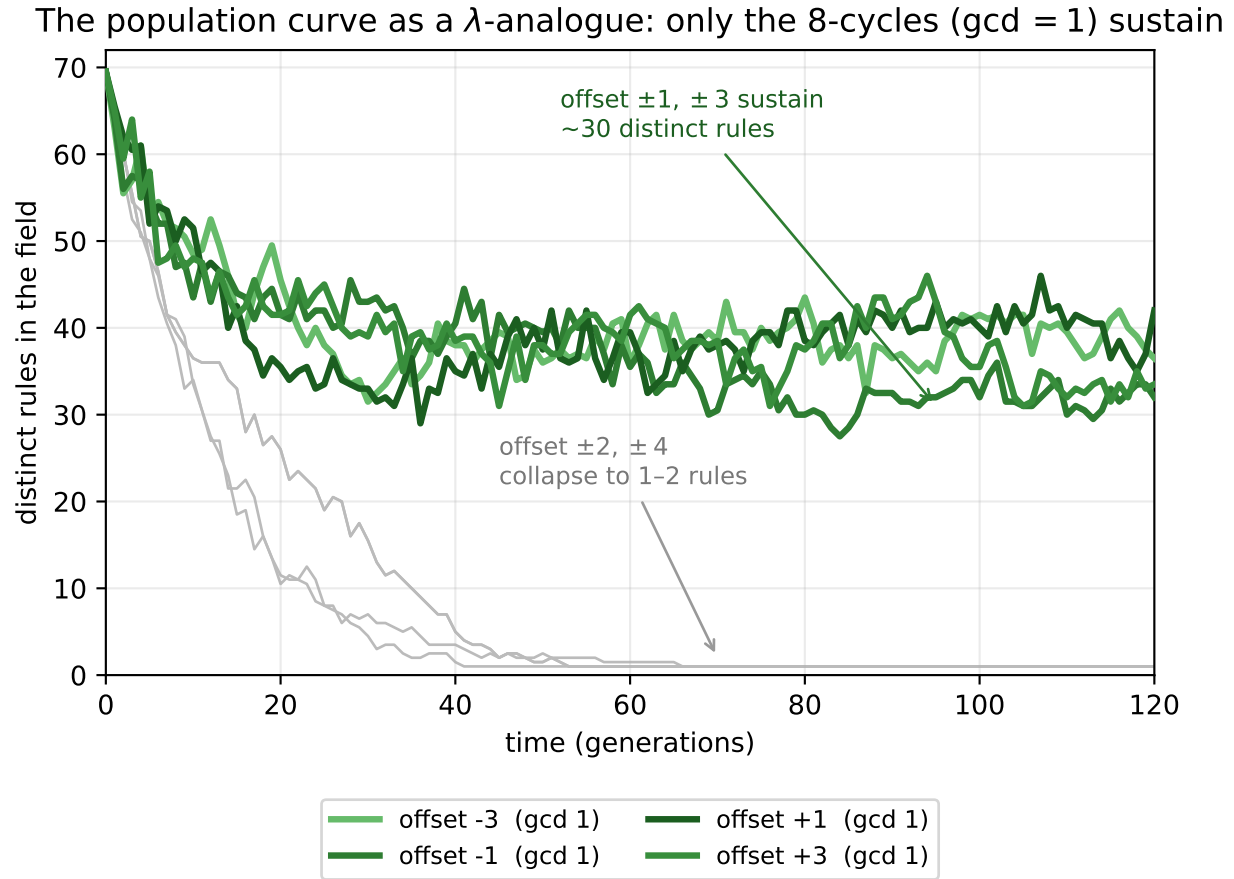


Figure 5

The parity dichotomy, by example (the census of all 40,320 permutations is summarised in the text). Five operators as genotype (top) and phenotype (bottom) over time. Left: both all-even operators possess fixed bytes (Theorem 1), but only rotate+2 settles within the run, reaching its $2^k = 2^2$ fixed rules; the single-8-cycle rotate+1 remains active at diversity 12. Right: the odd-cycle operators (reduced rotate+2, acting on a subset so the remaining positions are odd one-cycles) possess no fixed byte and stay diverse in these runs. All five use stochastic elementary writes without neighbour blending. The figure therefore illustrates the theorem's fixed-byte classification while also showing that all-even parity makes settling possible, not inevitable on a finite horizon.

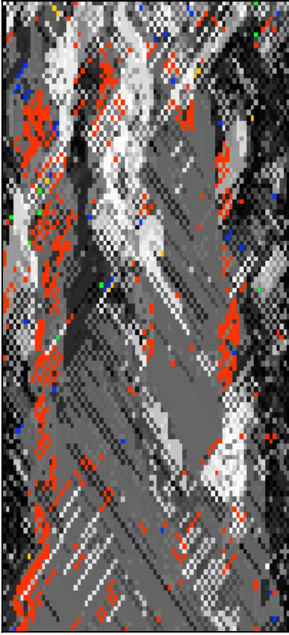
**Figure 6**

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70 ; the four 8-cycles, $\gcd(\text{offset}, 8) = 1$ (offset ± 1 and ± 3 , green), keep a large diverse population, while offset ± 2 and ± 4 (grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure S3.1. Figure 5 gives the corresponding without-blend comparison for offsets $+1$ and $+2$: the former remains active at lower diversity over that run, while the latter settles onto its four fixed rules.

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[6 7 0 2 1 4 3 5] — **two 4-cycles**

sustains (live / live)
genotype

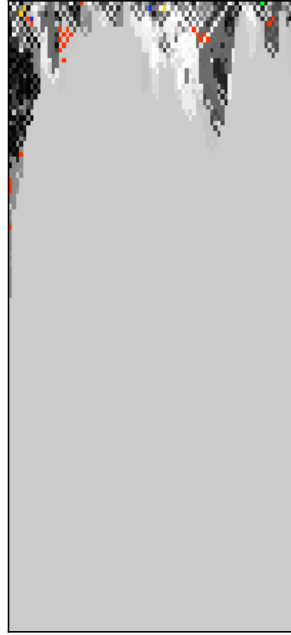


phenotype



[2 3 4 5 6 7 0 1] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



phenotype



Figure 7

Cycle structure does not determine aliveness. Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left, [6 7 0 2 1 4 3 5] sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the historical offset+2 operator (its conjugate). Right, the Wolfram-order rotation offset+2 = [2 3 4 5 6 7 0 1] has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.

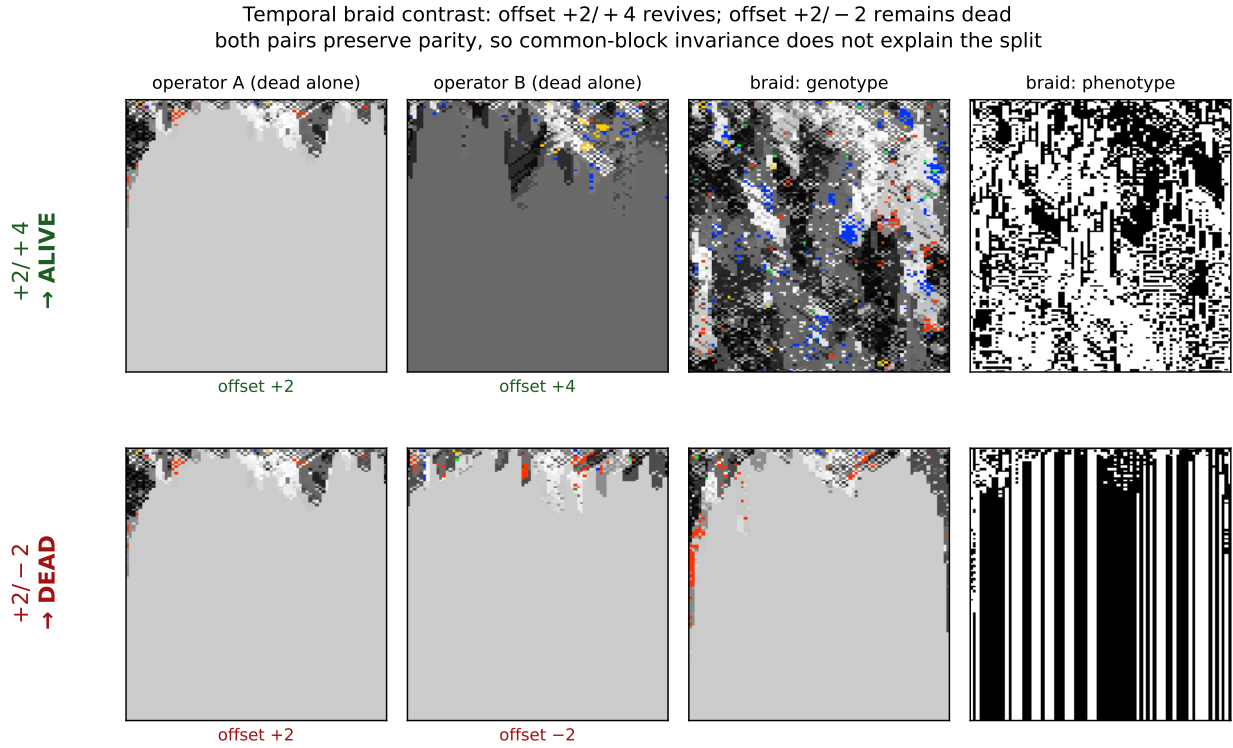


Figure 8

***Temporal braiding can sustain aliveness.** Two operators that each collapse alone are braided by alternating their elementary spatial updates. Top: offset+2 (two 4-cycles) and offset+4 (four 2-cycles) are each genotypically dead, yet their braid is alive (genotype change-rate 0.76, vs. 0 for either alone). Bottom: offset+2 braided with offset-2 stays dead. Both tested pairs preserve parity, so the contrast is not explained by the existence of that common invariant partition. The result concerns time-dependent coupled dynamics, not composition of the simultaneous maps P_σ .*

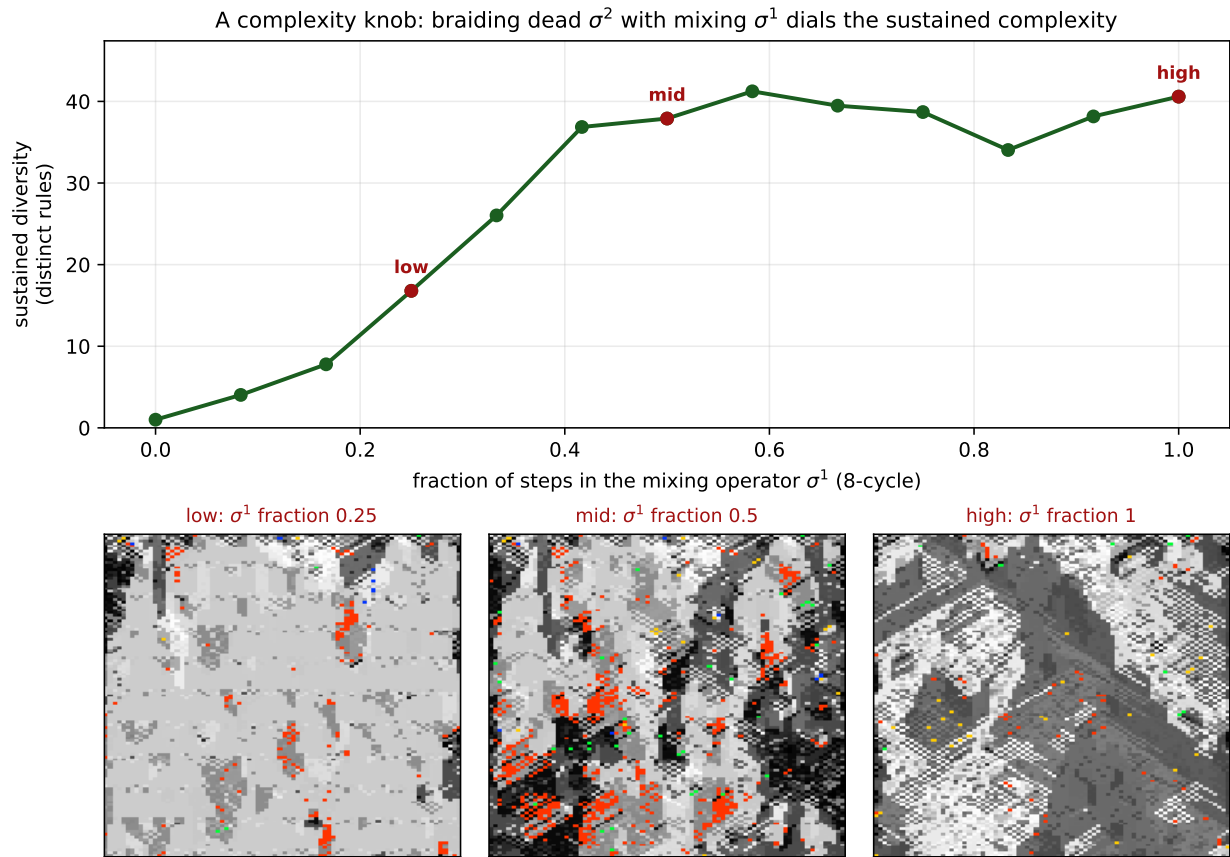


Figure 9

A diversity-control schedule. Periodic 12-step schedules mix the sustaining offset+1 operator σ^1 with the collapsing offset+2 operator σ^2 . Mean late-window distinct-rule diversity (five seeds) rises from 1.0 with no σ^1 steps to a high-diversity plateau once its fraction reaches roughly 0.4. The three fields show fractions 0.25, 0.50, and 1.0 at the same representative seed.

offset+1 feedforward scan across system size L (row width in cells, 30–240; 32 seeds): a smooth crossover with no critical point, and a susceptibility that peaks in the band but drifts and barely grows—metastability, not criticality (bottom: one realization across q , $L = 240$)

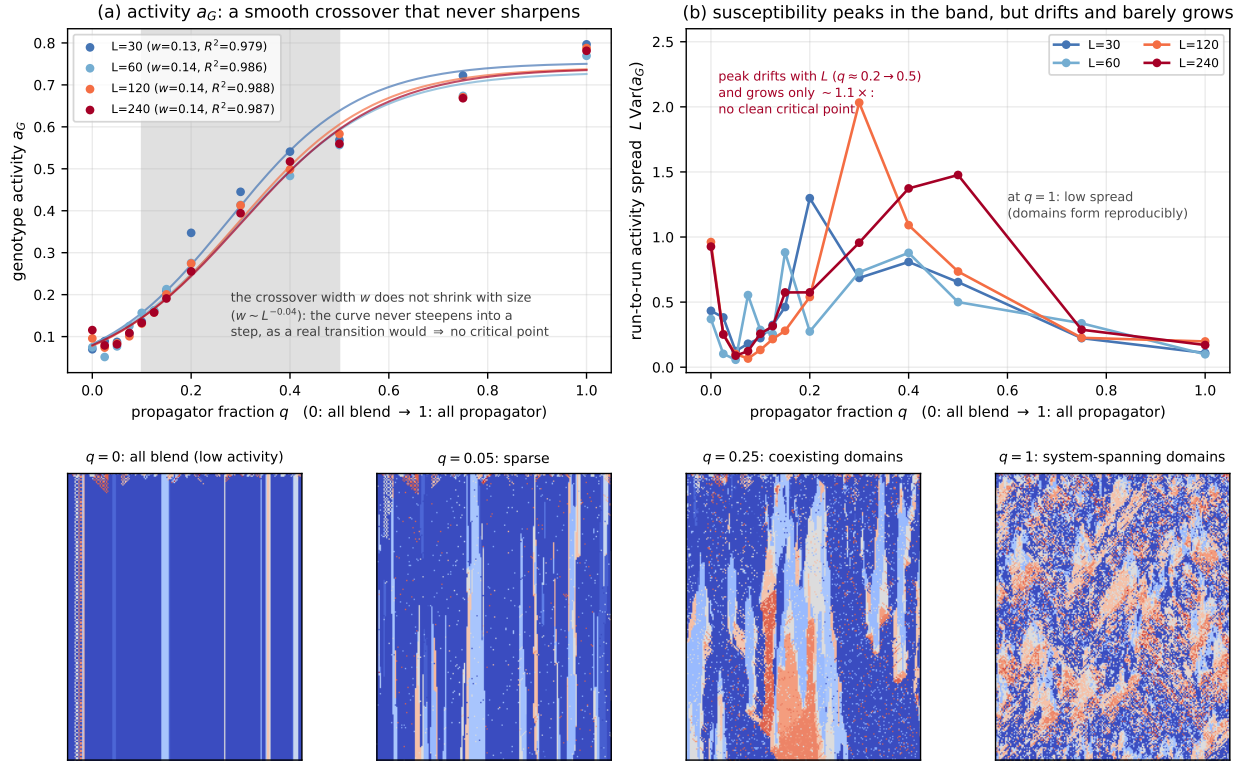


Figure 10

Finite-size scan across the propagator fraction. Offset +1 feedforward, $L = 30$ –240, 32 seeds per size. (a) genotype activity with logistic fits; (b) size-scaled spread as a susceptibility proxy; (c) finite-horizon collapse frequency; (d) the Binder cumulant. Bottom: one paired realisation at $L = 240$ tinted by isolated-rule activity.

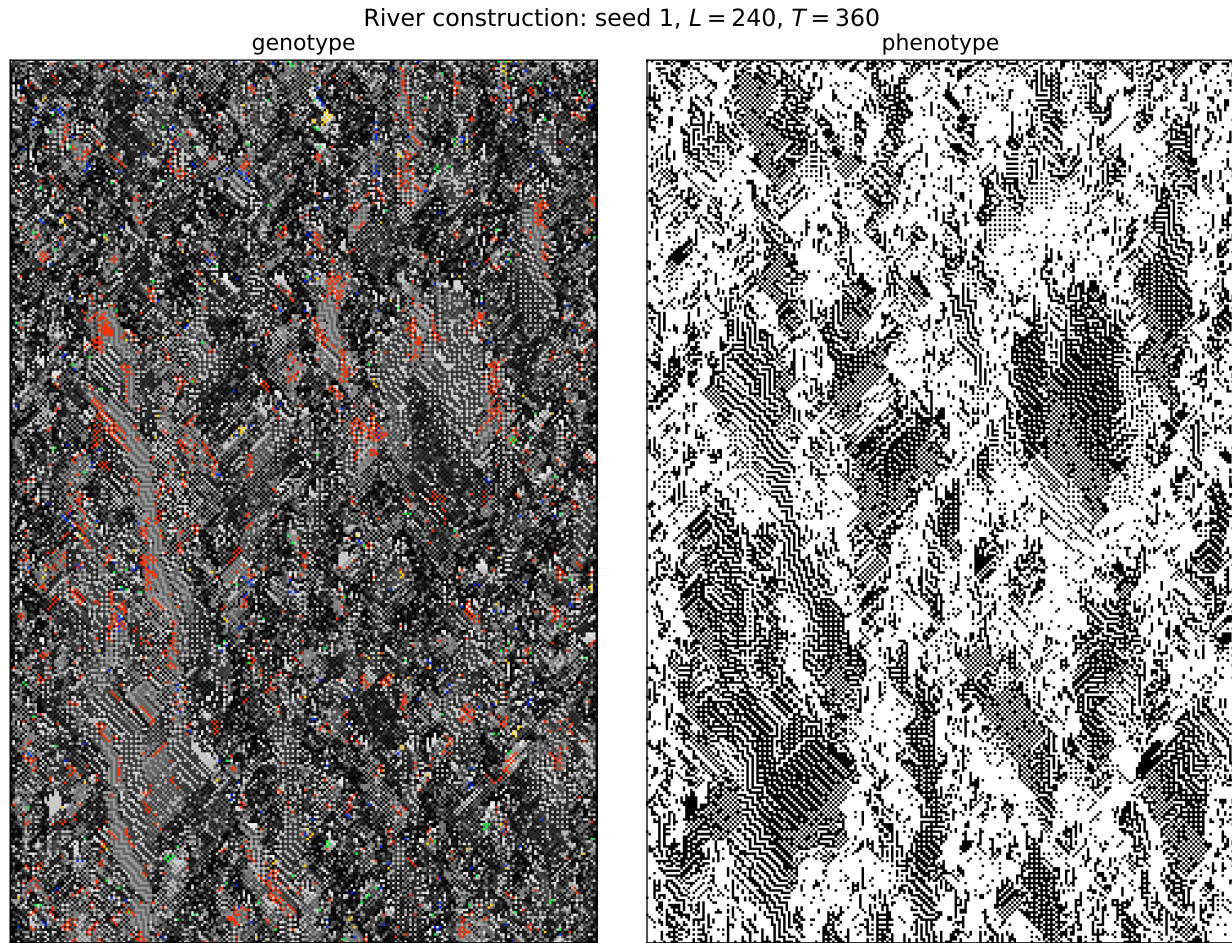


Figure 11

A larger river spacetime plate. One pinned representative run (seed 1) at $L = 240$, $T = 360$, with genotype at left and phenotype at right. The offset-+2 propagator is composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule). Alternating phenotype domains branch, merge, and reorganise throughout the run rather than resolving to a homogeneous or short-period field. Genotype is coloured by rule as in Figures 2–Figure S3.1: greyscale by byte value, with well-known ECAs highlighted and red denoting Rule 110. Where the bare rotations collapse or cycle (Figure S3.1), this composition sustains structured phenotype for the length of the observed run.

A single phenotype-bit flip spreads farther in the river than in the feedback-cut control
(same seed, same RNG tape, same construction; single-seed sweep over all 80 sites, $L=80$)

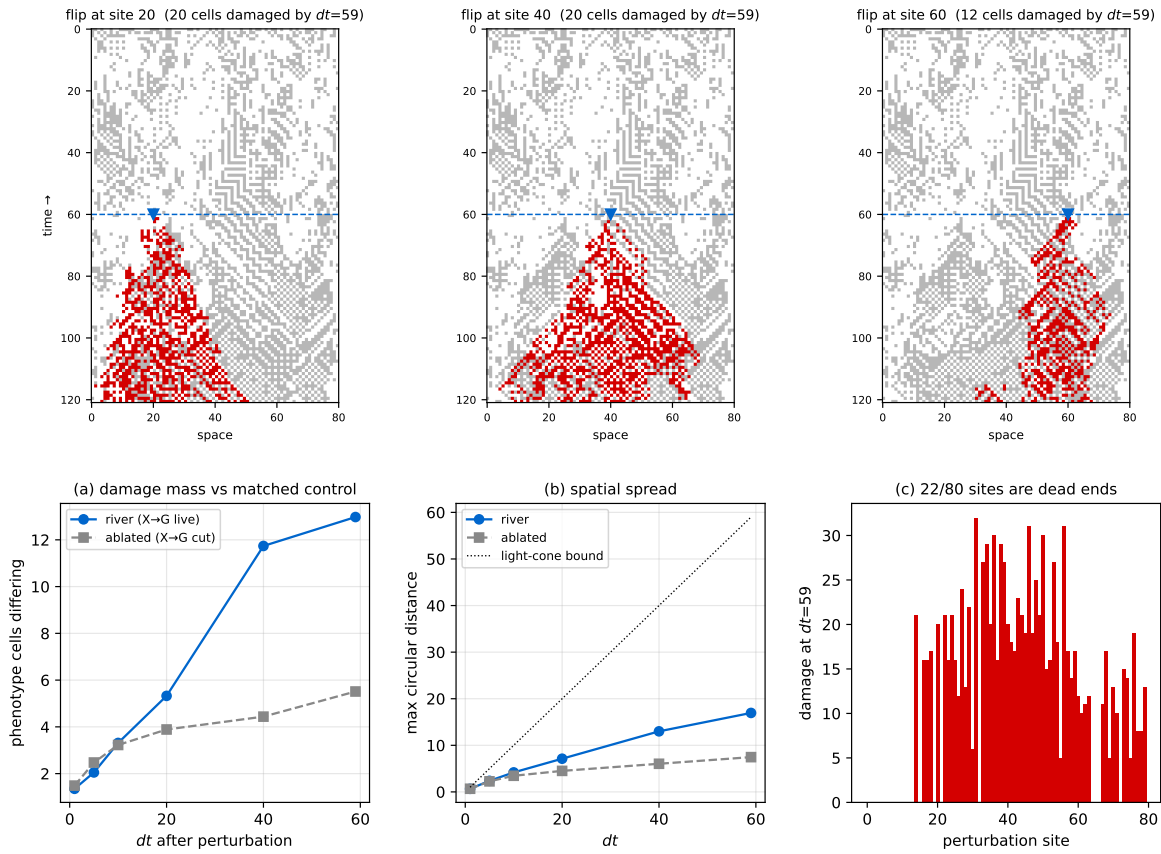


Figure 12

Damage from a single phenotype flip. Live construction against its matched control, $L = 80$, $t^ = 60$, swept over all 80 sites; mass and spread at a sequence of horizons dt .*

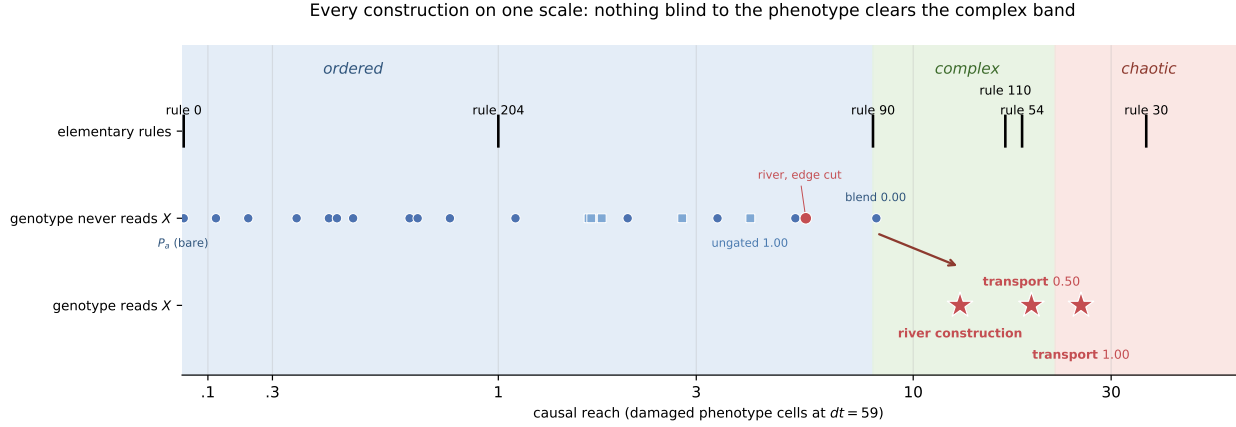


Figure 13

Every construction on one scale. Causal reach under the protocol of Supplement 1, Box 9, with regime bands drawn from the elementary rules. Rows divide constructions by whether the genotype update reads the phenotype.

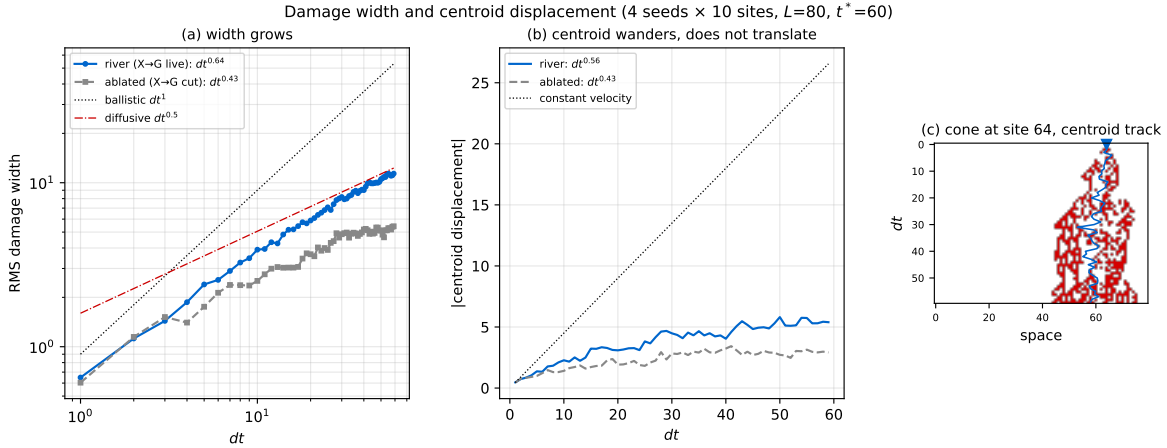
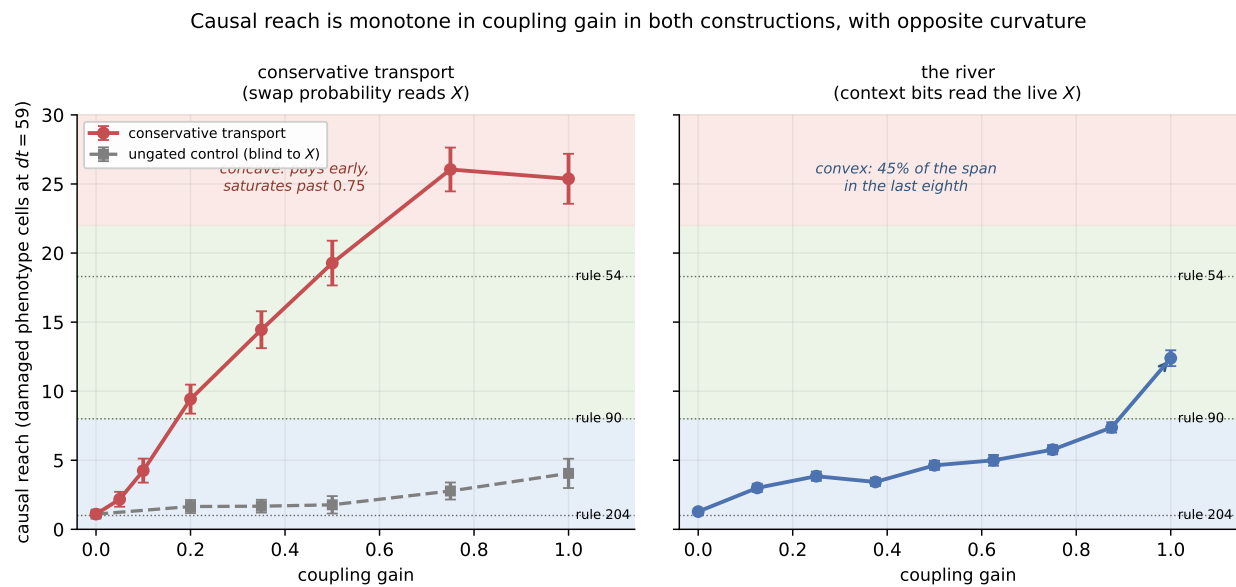


Figure 14

Damage width and centroid displacement. (a) damage width against dt , log-log, with ballistic and diffusive references; (b) centroid displacement; (c) a single cone with the centroid track overlaid. The packet disperses while its centroid stays near the perturbation site. We do not read this as excluding particle-like transport: under the same protocol rule 110, whose gliders support universal computation, gives $dt^{0.68}$ and $dt^{0.73}$, so a random single-site perturbation does not produce a fixed-width translating packet even where gliders certainly exist.

**Figure 15**

***Two couplings, dialled.** Causal reach against coupling gain for conservative transport (left, with an ungated control at matched mean swap probability) and for the river (right). Error bars are standard errors.*

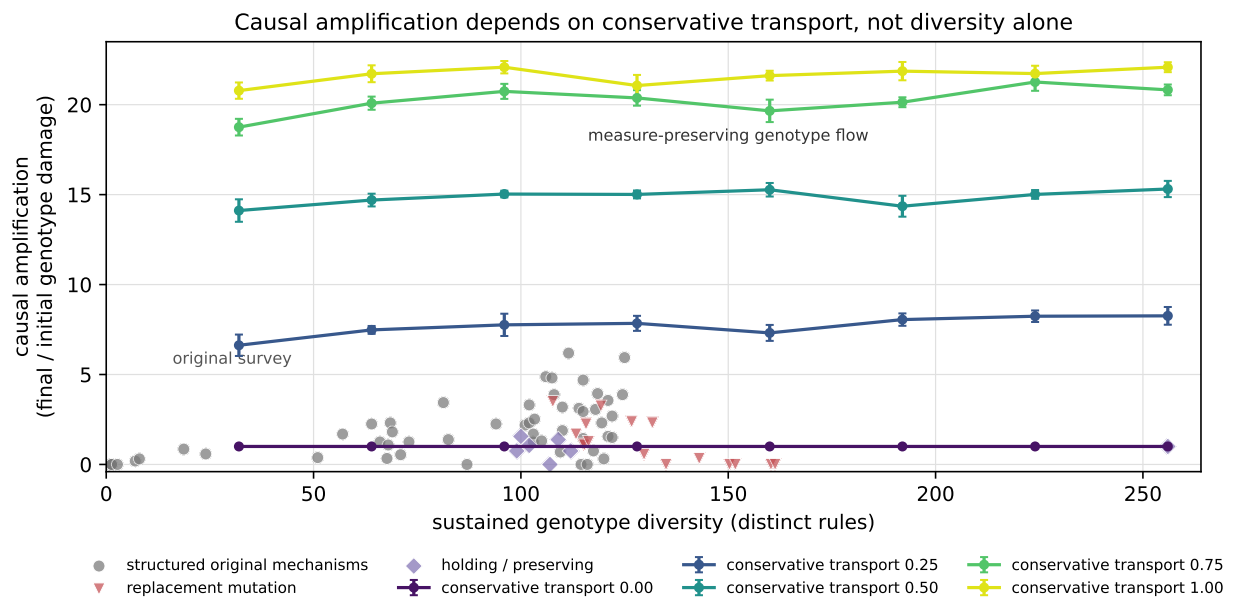


Figure 16

Causal reach against sustained genotype diversity. The original 76 configurations, grouped by mechanism, with the conservative-transport runs overlaid as one band per transport rate.

References

- Anthropic. (2026a). *Claude Opus*. Retrieved July 25, 2026, from <https://www.anthropic.com/claude/opus>
- Anthropic. (2026b). *Introducing Claude Fable 5*. Retrieved July 27, 2026, from <https://platform.claude.com/docs/en/about-claude/models/introducing-claude-fable-5>
- Aracena, J. (2008). Maximum number of fixed points in regulatory Boolean networks. *Bulletin of Mathematical Biology*, 70(5), 1398–1409.
- Corneli, J., & Maclean, E. (2015). The search for computational intelligence [v1].
- Crutchfield, J. P., & Hanson, J. E. (1993). Turbulent pattern bases for cellular automata. *Physica D: Nonlinear Phenomena*, 69, 279–301.
- Crutchfield, J. P., & Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2), 105–108. <https://doi.org/10.1103/PhysRevLett.63.105>
- Denmunzio, A., Formenti, E., & Provillard, J. (2011). Non-uniform cellular automata: Classes, dynamics, and decidability [arXiv:1107.5228].
- Hanson, J. E., & Crutchfield, J. P. (1997). Computational mechanics of cellular automata: An example. *Physica D: Nonlinear Phenomena*, 103, 169–189.
- Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3), 12–37. [https://doi.org/10.1016/0167-2789\(90\)90064-V](https://doi.org/10.1016/0167-2789(90)90064-V)
- Lizier, J. T., Prokopenko, M., & Zomaya, A. Y. (2008). Local information transfer as a spatiotemporal filter for complex systems. *Physical Review E*, 77(2), 026110.
- Mitchell, M., Hraber, P. T., & Crutchfield, J. P. (1993). Revisiting the edge of chaos: Evolving cellular automata to perform computations. *Complex Systems*, 7(2), 89–130.
- Mori, T., Kudo, K., Namagawa, Y., Nakamura, R., Yamakawa, O., Suzuki, H., & Uesugi, T. (1998). Edge of chaos in rule-changing cellular automata. *Physica D: Nonlinear Phenomena*, 116, 275–282.

- OEIS Foundation Inc. (2011). Sequence A001818: Squares of double factorials; number of permutations of S_{2n} in which all cycles have even length [<https://oeis.org/A001818>; even-cycle interpretation dated 2011].
- OpenAI. (2026). *Codex*. Retrieved July 25, 2026, from <https://openai.com/codex/>
- Packard, N. H. (1988). Adaptation toward the edge of chaos. In J. A. S. Kelso, A. J. Mandell, & M. F. Shlesinger (Eds.), *Dynamic patterns in complex systems* (pp. 293–301). World Scientific.
- Pavlic, T. P., Adams, A. M., Davies, P. C. W., & Walker, S. I. (2014). Self-referencing cellular automata: A model of the evolution of information control in biological systems [arXiv:1405.4070].
- Riordan, J. (1968). *Combinatorial identities*. Wiley.
- Sakai, S., & Kanno, M. (2002). Structure of rule table and phase diagram of one dimensional cellular automata [arXiv:nlin/0204045].
- Schaller, M., & Svozil, K. (2025). Irreducible rules and equivalence classes of one-dimensional cellular automata [arXiv:2512.08117].
- Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4), 817–879. <https://doi.org/10.1023/A:1010388907793>
- Stanley, R. P. (1999). *Enumerative combinatorics, volume 2* [Problem 5.34(c)]. Cambridge University Press.
- Wuensche, A. (1999). Classifying cellular automata automatically: Finding gliders, filtering, and relating space-time patterns, attractor basins, and the Z parameter. *Complexity*, 4(3), 47–66.

Supplement 1: Empirical Findings

Notebook

How to read this supplement

The boxes below occur in the same order as the corresponding results in the main paper. References to figures, tables, theorems, and sections without a “Supplement” prefix refer to the main paper. The source file `supplement1-findings.tex` is canonical: the main paper contains pointers, not duplicate finding text.

Empirical finding 1: The full permutation census

Every permutation containing an odd cycle lacks a fixed byte (Supplement 2, Theorem 1); under the census protocol, all 29,295 such operators were classified genotype-alive. The theorem excludes a common fixed byte for all elementary writes; the observed 100% alive rate is an additional dynamical result. Every all-even cycle type splits between live and dead operators. Among those types, the alive fraction falls as the number of cycles k increases: $[8]$ ($k=1$) is 83% alive, $[2\,6]$ and $[4\,4]$ ($k=2$) 70/67%, $[2\,2\,4]$ ($k=3$) 49%, and $[2\,2\,2\,2]$ ($k=4$) 26%. Overall, 92.5% have a live genotype. The regime split is 82.7% live/live, 9.8% live/dead, 6.0% dead/dead, and 1.4% dead/live, so all four regimes occur among bijective operators.

Empirical finding 2: The rotation survey

Within the wrap-rotation subfamily, every non-trivial rotation is all-even and so possesses fixed rules (Supplement 2, Theorem 1), yet under the neighbour-blend coupling the tested rotations split by cycle structure (Supplement 3, Figure 1 and Figure 6):

- the single 8-cycles— $\gcd(\text{offset}, 8) = 1$, $\text{offset} \pm 1$ and ± 3 —*sustain* a high-diversity field, holding ~ 30 – 40 distinct rules through the longest runs we made (400 generations);

- offset ± 2 (gcd 2, two 4-cycles) and offset ± 4 (gcd 4, four 2-cycles) *collapse* to one or two rules within a few dozen generations.

The distinct-rule population curve (Figure 6) separates the two by more than an order of magnitude in a single scalar. Possessing a fixed rule and having the coupled field stay near it are different properties. The present claim is limited to two seeds per rotation and treats population size as a descriptive observable, not as an order parameter in the statistical-mechanics sense.

Empirical finding 3: A braid of two collapsing operators can sustain the field

At width 80, horizon 120, and seed 0, offset +2 and offset +4 each have genotype change rate 0 when held alone, while their alternating braid has change rate 0.76 (Figure 8, top). The matched schedule control offset +2/offset -2 remains dead (Figure 8, bottom). This establishes that two collapsing components can construct a sustaining schedule. It does not establish the previously suggested block-system criterion: all three even offsets preserve the evens/odds partition, and both pairs generate the same translation subgroup $\langle +2 \rangle$. The cause of the contrasting braid outcomes remains to be identified.

Empirical finding 4: The mixing fraction is a reproducible diversity control

Let σ be displayed-position offset +1. We ran periodic 12-step schedules mixing the sustaining 8-cycle σ^1 with the collapsing two-4-cycle operator σ^2 , at width 80 for 120 generations and five seeds per mixture. Mean late-window diversity rises from 1.0 rule when the σ^1 fraction is zero, through 16.8 at fraction 0.25, to 36.9 at fraction 5/12; above that it remains in the range 34–41 rather than increasing monotonically (Figure 9). The schedule fraction is thus a tunable control for entry into a sustained high-diversity regime, with saturation rather than a continuously linear response.

Empirical finding 5: The rotation split occurs with and without neighbour blending

Figure 5 uses stochastic elementary writes without neighbour blending; Supplement 3, Figure 1 and Figure 6 include the blend. Over the displayed horizons, offset +1 remains active and offset +2 settles under both protocols, although their diversity levels and transients differ. These runs therefore do not identify neighbour blending as the cause of the split; they show that fixed-byte existence alone does not determine finite-horizon settling under either protocol.

Empirical finding 6: No finite-size evidence of a critical point over the tested range

The natural control parameter is the *propagator fraction* q ($q = 0$ all blend, $q = 1$ all propagator, offset +2). If the sustained phase sat at a critical point in q , a finite-size scan would show three things as the lattice grows: the transition in genotype activity $a_G(q)$ *sharpening*, its width shrinking toward zero; a susceptibility proxy peaking at a *fixed* interior value of q , the peak growing without bound; and Binder cumulants for different L crossing at that same value. Across widths $L = 30\text{--}240$ (32 seeds each) we observe none of the three.

First, $a_G(q)$ rises *smoothly*, well described by a logistic crossover ($R^2 > 0.99$, Figure 10a), and its width does not shrink over the tested sizes ($w \approx 0.13$, $w \sim L^{-0.04}$)—the transition is not sharpening. Second, the size-scaled run-to-run spread $L \text{Var}(a_G)$ (Figure 10b), used as a finite-size susceptibility proxy, does *peak* inside the crossover band and does grow ($\sim 5.7\times$ from the smallest width to the largest), but its location drifts rather than converging ($q = 0.75, 0.50, 1.00, 1.00$ as L increases), and for the two largest sizes the maximum lies at the sampled boundary $q = 1$, so no interior peak is resolved at all. Third, there is no Binder-cumulant crossing. The scan therefore provides no finite-size evidence for a sharpening transition over $L = 30\text{--}240$. These results support a

broad crossover, plausibly influenced by seed-dependent residence times, rather than a confirmed critical point; they do not exclude different scaling outside the tested range.

Empirical finding 7: The river sustains structured phenotype

In all six tested seeds at $L = 80$, $T = 120$, the river produces coherent bands of alternating phenotype that drift through a disordered background. In the larger pinned run at $L = 240$, $T = 360$ (Figure 11), those structures persist while branching, merging, and reorganising across the full spacetime field. This is a bounded six-seed observation plus one larger representative run, not a claim that the structure is an attractor or survives arbitrarily long.

Empirical finding 8: Phenotype-to-genotype feedback roughly doubles causal reach

All runs use $L = 80$, $T = 120$, $t^* = 60$, with damage measured on the *phenotype* layer as the number of cells differing between branches and distance as the shorter circular displacement. At $dt \leq 10$ the two branches are empirically indistinguishable. By $dt = 59$, in a single-seed sweep over all 80 sites, the live construction has damaged 12.97 cells against the control's 5.51 and spread 16.95 against 7.46 (Figure 12 plots that sweep). Replication uses a separate design—six seeds, 20 evenly spaced sites each, aggregated to one mean per seed so that the *seed* is the unit of replication. The live construction exceeds its control in 6/6 seeds, per-seed differences +1.90, +7.95, +6.65, +6.60, +6.05, +0.85 cells, mean +5.00 (sd 2.90). Because $n = 6$ makes a t -statistic fragile we quote the exact sign-flip permutation test over all 2^6 assignments: the observed mean is the extremum, one-sided $p = 1/64 \approx 0.016$ (paired $t(5) = 4.23$ for comparison). Perturbations do not spread uniformly: in the single-seed sweep 22 of 80 sites are dead ends, and they form a contiguous region.

One qualification on what the control cuts. `run-river-ablated-from` reads a phenotype frozen at the start of each run segment, so it cuts the *dynamic* edge but still presents

the genotype with a static field that, in the perturbed branch, already contains the flip. The 5.51 is therefore feedback-frozen, not feedback-absent; with the frozen reference supplied externally and identical in both branches, reach falls further, to 1.28 (Box 11). The doubling quoted here is the effect of making the read *current*, which is the smaller half of the total coupling effect.

Empirical finding 9: Causal reach on a scale calibrated against elementary rules

On a third design—four seeds, ten sites each, damage recorded at every dt —width grows as $dt^{0.64}$ for the live construction against $dt^{0.43}$ for its control, with centroid displacement scaling as $dt^{0.56}$ against $dt^{0.43}$. Exponents are least-squares fits over $dt \in [5, 59]$ on site-averaged curves, conditional on damage existing—a run whose perturbation has died contributes no width and is excluded rather than entered as zero—and we quote no interval on them (Figure 14); these are reported descriptively and we draw no inference from them about gliders or particles, for a reason given below. Placed on one scale against elementary rules run under an identical protocol (mean damaged cells at $dt = 59$): rule 0 0.00; rule 204 1.00; rule 90 8.00; **the construction** 12.97; rule 54 15.05; rule 110 20.23; rule 30 37.50. The instrument returns exactly zero where nothing propagates, and the construction propagates at a magnitude comparable to the elementary rules that do.

Empirical finding 10: The order parameter is the strength of the phenotype-to-genotype coupling

Run under the protocol above— $L = 80$, $t^* = 60$, one phenotype flip, damaged phenotype cells at $dt = 59$ —the constructions separate by *coupling*, not by which parameter they vary. Where the genotype update never reads the phenotype, nothing clears the bottom of the complex band: the bare operators (P_a 0.03, rot+1 2.05), braided writings (0.12, 5.20), spatial niches (0.50, 0.72), asynchronous refuges (0.47, 0.75), replacement

mutation (0.23, 0.85), the preserving limit (1.10), and blend strength across its whole range—1.00 giving 0.03, 0.70 giving 0.55, 0.35 giving 3.38, and 0.00, the most propagating feedforward configuration we found, giving 8.15, level with rule 90. Most sit below rule 204 (1.00), the frozen identity.

The coupling is not merely present or absent; it is a dial, and reach is graded and monotone along it. Conservative transport whose swap probability is gated on the local phenotype interface rises from 1.10 at rate 0 through 4.25 at 0.10, 9.43 at 0.20, 14.45 at 0.35, 19.27 at 0.50 and 26.05 at 0.75, crossing rule 90 between rates 0.10 and 0.20 and passing rules 110 and 54 before saturating. The river construction sits at 12.97 on the same scale, and cutting its single edge returns it to 5.51, back among the feedforward constructions.

That this is the *coupling* and not merely rule mobility is established by an ungated control. Replacing the phenotype-gated swap probability with a constant one leaves the mean number of swaps unchanged—for a balanced field the gated mean is exactly the rate—and leaves the dynamics bijective and the rule histogram invariant, so the two differ only in whether the genotype can see the phenotype. The ungated control never leaves the ordered band: 1.65, 1.68, 1.77, 2.77 and 4.05 at rates 0.20 through 1.00, against 9.43 to 25.38 for the gated version, a factor of 5.7 to 10.9 at matched mobility (Figure 13).

An order parameter therefore exists, and the classical picture is recovered in it: reach is a graded, monotone function of coupling strength with a locatable crossing into the complex band. What is unusual is only *which* coordinate it is. Every parameter internal to the update— λ , propagator fraction, sustained diversity, blend strength, mutation rate, refuge probability, niche width, and rule mobility itself—is silent or nearly so. The coordinate that governs is the gain of the loop from the phenotype back to the genotype, which conventional scans do not sweep because architecture is usually treated

as a property a model has or lacks rather than as a quantity.

Empirical finding 11: Coupling is a dial, not a switch, and reach is monotone along it

Both constructions that read the phenotype admit a gain parameter, and in both causal reach is graded and monotone in it under the protocol of Box 9.

Figure 15 plots both dials. For conservative transport the gain is the phenotype-gated swap rate. Reach rises 1.10, 2.17, 4.25, 9.43, 14.45, 19.27, 26.05, 25.38 at rates 0, 0.05, 0.10, 0.20, 0.35, 0.50, 0.75, 1.00: it crosses rule 90 between rates 0.10 and 0.20, passes rules 110 and 54, and saturates past 0.75.

For the river the gain is the *currency* of the four context bits the genotype step reads. Each cell at each step reads the live phenotype with probability γ and a frozen reference otherwise, so γ is the fraction of the genotype’s view of the phenotype that is causally current. Over three seeds and eighty sites, reach rises 1.28, 3.00, 3.85, 3.43, 4.63, 5.00, 5.77, 7.38, 12.39 across $\gamma = 0$ to 1 in eighths, monotone within one standard error, with standard errors 0.19 to 0.57.

Three properties make the river dial an instrument rather than a knob. It is anchored at both ends: $\gamma = 1$ reproduces the live river at 12.97 exactly. The random tape is preserved—the propagator draw is consumed once per cell at every γ , with gate coins from a separate stream re-seeded identically in both branches, so the gate injects no divergence of its own. And because the frozen field is a real phenotype, marginals and spatial structure are matched at every γ ; only causal currency varies.

The two curves have opposite curvature. Transport is concave, buying most of its reach in the first fifth of the dial. The river is convex: 45% of its span arrives in the last eighth, and only $\gamma = 1$ clears rule 90, so one stale read in eight costs 40% of the reach. A plausible reading is that the river’s first-match fires on a four-bit conjunction, so a stale context does not degrade the match but breaks it; we have not tested this, and

gating the four bits independently rather than together would separate it.

A one-seed pilot showed a non-monotonic dip near $\gamma = 0.5$ that did not survive three seeds. We record it because we nearly believed it.

Empirical finding 12: Sustained diversity does not determine causal reach; conservative transport breaks the apparent interior optimum

The original 76 configurations produce an apparent rise and fall: binned medians rise from 0.10 cells below 40 sustained rules to 2.34 at 125–135, then fall to zero above 135. That descent is completely confounded. Every original configuration above 135 obtains its diversity either by replacement mutation or by holding the genotype fixed. The observed failures are real—mutation erases the lesion and holding prevents its spread—but they do not show that high diversity suppresses propagation.

We therefore add a conservative eighth mechanism. Each run begins from a balanced empirical rule measure with support 32, 64, 96, 128, 160, 192, 224, or 256. Alternating layers of disjoint local pairs exchange their rules with a probability controlled by a transport rate and the local phenotype interface. These exchanges are bijections: no rule is created or destroyed, and the complete genotype histogram and its Shannon diversity remain invariant at every checked step. A local transposition supplies the perturbation while preserving that same measure in both branches. We run five transport rates, six seeds, and eight perturbation sites for 40 response steps, giving 1,920 paired responses. Figure 16 reports causal amplification—final genotype damage divided by the two initially exchanged sites—so the conservative responses share a scale with the original single-site survey. With transport zero, amplification is exactly 1.00 at every diversity. At transport rates 0.25, 0.50, 0.75, and 1.00, the ranges across the full diversity axis are respectively 6.63–8.26, 14.11–15.31, 18.75–21.26, and 20.78–22.08. There is no descending limb: within each transport rate the dependence on diversity is approximately flat or slightly positive, while changing transport changes reach by more than an order of

magnitude.

Sustained rule diversity is therefore not an order parameter for causal reach. Mutation and holding are distinct suppression mechanisms, but they are not flanks of a common diversity optimum. Structured genotype transport is the control exposed by this experiment.

References

- Anthropic. (2026a). *Claude Opus*. Retrieved July 25, 2026, from <https://www.anthropic.com/claude/opus>
- Anthropic. (2026b). *Introducing Claude Fable 5*. Retrieved July 27, 2026, from <https://platform.claude.com/docs/en/about-claude/models/introducing-claude-fable-5>
- Aracena, J. (2008). Maximum number of fixed points in regulatory Boolean networks. *Bulletin of Mathematical Biology*, 70(5), 1398–1409.
- Corneli, J., & Maclean, E. (2015). The search for computational intelligence [v1].
- Crutchfield, J. P., & Hanson, J. E. (1993). Turbulent pattern bases for cellular automata. *Physica D: Nonlinear Phenomena*, 69, 279–301.
- Crutchfield, J. P., & Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2), 105–108. <https://doi.org/10.1103/PhysRevLett.63.105>
- Denmunzio, A., Formenti, E., & Provillard, J. (2011). Non-uniform cellular automata: Classes, dynamics, and decidability [arXiv:1107.5228].
- Hanson, J. E., & Crutchfield, J. P. (1997). Computational mechanics of cellular automata: An example. *Physica D: Nonlinear Phenomena*, 103, 169–189.
- Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3), 12–37. [https://doi.org/10.1016/0167-2789\(90\)90064-V](https://doi.org/10.1016/0167-2789(90)90064-V)
- Lizier, J. T., Prokopenko, M., & Zomaya, A. Y. (2008). Local information transfer as a spatiotemporal filter for complex systems. *Physical Review E*, 77(2), 026110.
- Mitchell, M., Hraber, P. T., & Crutchfield, J. P. (1993). Revisiting the edge of chaos: Evolving cellular automata to perform computations. *Complex Systems*, 7(2), 89–130.
- Mori, T., Kudo, K., Namagawa, Y., Nakamura, R., Yamakawa, O., Suzuki, H., & Uesugi, T. (1998). Edge of chaos in rule-changing cellular automata. *Physica D: Nonlinear Phenomena*, 116, 275–282.

- OEIS Foundation Inc. (2011). Sequence A001818: Squares of double factorials; number of permutations of S_{2n} in which all cycles have even length [<https://oeis.org/A001818>; even-cycle interpretation dated 2011].
- OpenAI. (2026). *Codex*. Retrieved July 25, 2026, from <https://openai.com/codex/>
- Packard, N. H. (1988). Adaptation toward the edge of chaos. In J. A. S. Kelso, A. J. Mandell, & M. F. Shlesinger (Eds.), *Dynamic patterns in complex systems* (pp. 293–301). World Scientific.
- Pavlic, T. P., Adams, A. M., Davies, P. C. W., & Walker, S. I. (2014). Self-referencing cellular automata: A model of the evolution of information control in biological systems [arXiv:1405.4070].
- Riordan, J. (1968). *Combinatorial identities*. Wiley.
- Sakai, S., & Kanno, M. (2002). Structure of rule table and phase diagram of one dimensional cellular automata [arXiv:nlin/0204045].
- Schaller, M., & Svozil, K. (2025). Irreducible rules and equivalence classes of one-dimensional cellular automata [arXiv:2512.08117].
- Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4), 817–879. <https://doi.org/10.1023/A:1010388907793>
- Stanley, R. P. (1999). *Enumerative combinatorics, volume 2* [Problem 5.34(c)]. Cambridge University Press.
- Wuensche, A. (1999). Classifying cellular automata automatically: Finding gliders, filtering, and relating space-time patterns, attractor basins, and the Z parameter. *Complexity*, 4(3), 47–66.

Supplement 2: Definitions, Theorems and Worked Examples

This supplement holds the formal apparatus of the main paper: the definitions that fix the object, the two theorems and their corollaries that classify it, and the worked examples that carry a reader through one member of the family by hand. The main paper states each result in prose where it is used and points here for the statement, the proof, and the arithmetic. Numbering matches the order of first use in the paper.

Definition 1: Writings, elementary writes, and propagators

A *writing* is any function $w: \{0, \dots, 7\} \rightarrow \{0, \dots, 7\}$ on the truth-table positions. Its elementary update at source position p is

$$(U_{w,p}g)(q) = \begin{cases} \neg g(p), & q = w(p), \\ g(q), & q \neq w(p). \end{cases}$$

Thus one update complements source bit p and writes it to target $w(p)$. If w is a bijection—a permutation $\sigma \in S_8$ —all eight writes can be performed simultaneously without collisions, giving the *permutation-propagator* P_σ in Equation (1). Its fixed bytes are exactly the bytes fixed by every $U_{\sigma,p}$. For a non-injective writing, individual updates $U_{w,p}$ remain well defined, but a simultaneous P_w requires a collision-resolution convention; such a writing also has *gaps*, or positions targeted by no source.

Observation 1: Permutations form a small bijective core

Permutations are only the bijective core of a far larger class: there are $8! = 40,320$ permutations but $8^8 = 16,777,216$ writings, so the permutations are 0.24% of the whole. The cycle-structure algebra of *The Classification* is available only on this core. The historical operator $[00123456]$ (Figure 3) is itself a non-bijective writing—bits 0 and

1 both write to position 0, and position 7 is never written—and its behaviour ranges widely between its two standard-order forms.

Worked example 1: A simultaneous propagator by hand

Write an ECA rule in Wolfram’s standard neighbourhood order:

position	0	1	2	3	4	5	6	7
neighbourhood	111	110	101	100	011	010	001	000
Rule 110	0	1	1	0	1	1	1	0

Thus Rule 110 is the byte 01101110. Let $\sigma(p) = p+2 \pmod{8}$ on the displayed position indices: each source bit is complemented and written two positions further to the right, wrapping at the end. Carrying out the eight writes gives

$$P_{+2}(01101110) = 01100100,$$

which is Rule 100. This changes what the cell does to its black-and-white phenotype. For example, neighbourhood 011 maps to 1 under Rule 110 but to 0 under Rule 100. The propagator has not directly flipped a phenotype cell; it has changed the truth table that the cell will use.

Definition 2: Genotype and phenotype

Each cell carries an eight-bit rule. The field of rules—the *genotype*—governs a field of black-and-white states—the *phenotype*—and neighbouring rules also interact. One complete local update (Box 2) evolves both layers; this is the computation behind the space–time diagrams in Figure 2.

Worked example 2: One site, one complete MetaCA step

At time t , site x holds an eight-bit rule $g_x(t)$ and a phenotype bit $X_x(t)$. One ordinary coupled step has three parts.

1. **Express the current rule.** The phenotype neighbourhood is evaluated by the rule at its centre. With numerical neighbourhood value $n_x(t) = 4X_{x-1}(t) + 2X_x(t) + X_{x+1}(t)$,

its displayed position is $7 - n_x(t)$:

$$X_x(t+1) = g_x(t)(7 - n_x(t)).$$

2. **Blend neighbouring rules.** For each truth-table position j , inspect the three genotype bits $(g_{x-1}(j), g_x(j), g_{x+1}(j))$. If the two outer bits agree, copy their value. Thus $(0, 1, 0)$ blends to 0. If they disagree, the centre rule decides by treating the triple as one of its ECA neighbourhoods, converting its numerical value n to displayed position $7 - n$ as above. The eight coordinate-wise decisions form an intermediate rule

$$h_x = B(g_{x-1}(t), g_x(t), g_{x+1}(t)).$$

3. **Write one propagated bit.** Draw a source position $K_x(t) \in \{0, \dots, 7\}$ uniformly and apply its elementary write to the intermediate byte:

$$g_x(t+1) = U_{\sigma, K_x(t)}(h_x).$$

The ordinary spatial runs make one such elementary write per cell and generation. The historical Figure 8 variant made two. By contrast, P_σ denotes the simultaneous eight-write map classified in *The Classification*; its fixed bytes are precisely the bytes left unchanged by every possible elementary write.

Theorem 1

P_σ has a fixed point (a fixed rule) if and only if every cycle of σ has even length. Since a one-cycle has odd length, this already requires σ to be fixed-point-free.

Worked example 3: Even and odd cycles

For offset +2 on eight truth-table positions,

$$\sigma = (0\ 2\ 4\ 6)(1\ 3\ 5\ 7).$$

A fixed rule must change colour at every arrow because $g(\sigma k) = \neg g(k)$. Around either four-cycle it can therefore read 0101 or 1010: after four negations it returns consistently to its

starting bit. The two four-cycles are coloured independently, so two binary choices fix all eight truth-table entries $\text{bit}[0], \dots, \text{bit}[7]$. Read back as an ECA rule number—those eight bits in Wolfram neighbourhood order (the table of *The Object*) taken as an integer in 0–255—the four colourings are the fixed rules

$$2^2 = 4 : \quad 51, 102, 153, 204,$$

each with four zero-bits and four one-bits. An odd cycle admits no such colouring: after an odd number of negations the bit returns to its starting position demanding its own opposite, so no fixed rule exists. Admitting a fixed rule and settling onto one are distinct. An operator carrying an odd cycle has no fixed rule for its field to collapse onto, and its field correspondingly stays diverse (the any-odd reduced rotate+2, Figure 5, right); whether a fixed-point-bearing operator collapses onto one of its fixed rules is a separate, dynamical question (*Fixed Points Are Not Attractors*).

Corollary 1

An all-even σ with k cycles has exactly 2^k fixed bytes; any σ with an odd cycle has none.

Definition 3: Langton's activity λ

λ measures a rule's *activity*: the fraction of its neighbourhood entries that map to the non-quiescent state (Langton, 1990)—for an eight-bit ECA table, the Hamming weight over eight, $\lambda = \|r\|_1/8$, from 0 (every neighbourhood dies) through $1/2$ to 1. A rule with $\lambda = 1/2$ is *balanced*.

Observation 2: Balance is a truth-table property

Langton's thesis was that complex behaviour concentrates near a critical λ , shown most clearly for larger tables ($K = 4$, $N = 5$); for the two-symbol, three-neighbour ECAs Langton is explicit that λ only roughly correlates with dynamics, and later work is cautious (Mitchell et al., 1993). Theorem 2 below concerns the *value* $\lambda = 1/2$ itself—balance—which we keep separate from any empirical claim about what it predicts dy-

namically.

Theorem 2

Every fixed point of every operator in the family has exactly four one-bits; equivalently, Langton's activity parameter $\lambda = 4/8 = 1/2$.

Definition 4: Census protocol and aliveness

For each of the 40,320 permutations, we ran seeds 0–2 on a width-60 line for 100 generations. Initial rule bytes are independent uniform draws from 0–255, initial phenotype bits are independent fair binary draws, and both layers have fixed zero boundaries. Each genotype update uses the neighbour blend and one uniformly sampled elementary write described in Box 2. A layer's *change rate* is the mean fraction of cells differing between consecutive rows over the final 40 generations, averaged over the three seeds. The layer is *alive* when this rate is at least 0.10 and *dead* otherwise. Scoring genotype and phenotype separately gives four regimes: live/live, live/dead, dead/live, and dead/dead.

Worked example 4: Reading the rotation survey

The fields of Supplement 3, Figure 1 evolve under the *spatially coupled* dynamics of *Fixed Points Are Not Attractors*—stochastic elementary writes together with the coupling between cells—not P_σ applied cell-by-cell. This is what the survey turns on. Without the blend, Figure 5 already shows rotate+1 remaining active over the observed horizon while rotate+2 settles. With neighbour blending, the finite-horizon split persists and widens: the single 8-cycles— $\gcd(\text{offset}, 8) = 1$, offset ± 1 and ± 3 —sustain a high-diversity field, while offset ± 2 ($\gcd 2$) and ± 4 ($\gcd 4$) collapse; that split is what the survey shows.

The two panels are *almost* mirror images, and where they differ is instructive. Offset $+d$ and offset $-d$ are the same permutation exactly when $2d \equiv 0 \pmod{8}$ —so offset ± 4 (and the identity) are one operator with bit-identical panels, whereas offset ± 1 and ± 3 are distinct *inverse* rotations. Inverses share cycle structure, fixed rules, and the sustain-or-collapse verdict, but not their transient dynamics: offset $+1$ and offset -1 realise the same sustaining regime through visibly different fields.

Definition 5: Temporal braid

A *temporal braid* alternates two writings between complete coupled spatial updates: writing w_A is used at even times and w_B at odd times. More generally, a periodic *schedule* selects one writing at each time step. This preserves the intermediate spatial states and neighbour blends; it is not the algebraic composition $P_{\sigma_A} \circ P_{\sigma_B}$ of two simultaneous maps.

Definition 6: Interrupter

An *interrupter* is an update between elementary propagator writes that is not itself a propagator write. The coordinate-wise neighbour blend of Box 2 is the interrupter in the ordinary spatial runs; a hold gate that suppresses a scheduled write is another example. An interrupter changes the coupled dynamics but does not, by definition, guarantee sustained diversity.

Proposition 1: Composition parity

Products alternate between complemented and uncomplemented coordinate permutations. Each operator of Equation (1) negates every bit ($P_\sigma(g)[j] = \neg g[\sigma^{-1}(j)]$, a coordinate permutation followed by a global flip), so under composition the flips cancel in pairs: $P_\sigma \circ P_\tau$ is a *pure* coordinate permutation with no negation left—a bijection that fixes 2^c bytes for c cycles but carries none of the forced $\lambda = 1/2$ structure. An even number of propagators therefore composes to an uncomplemented coordinate permutation, and an odd number to another member of the permutation-propagator family.

Definition 7: River

A *river* is the composed operator obtained by following the collapsing offset-+2 propagator (gcd 2) with a phenotype-reading construction step—match the local phenotype against four template candidates, first match wins, and fall back to the all-zero rule when none matches.

Supplement 3: Additional Figures

Two survey galleries referred to from the main paper. Each shows a family of runs at a glance rather than carrying an argument on its own: the first sweeps the rotation offsets that Part I classifies, the second shows the spacetime fields behind the diversity mechanisms of Part II. They are collected here so the main narrative is not interrupted by four full pages of plates.

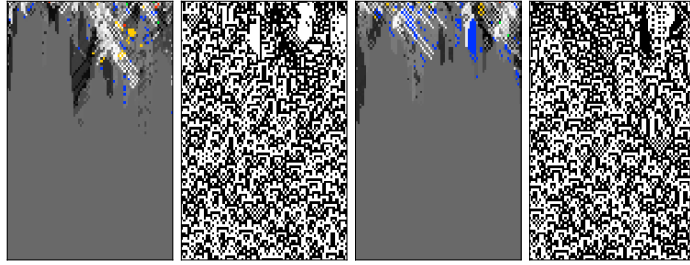
A family of operators, ranked by cycle structure (a · negative offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

offset -4

gcd 4 · four 2-cycles

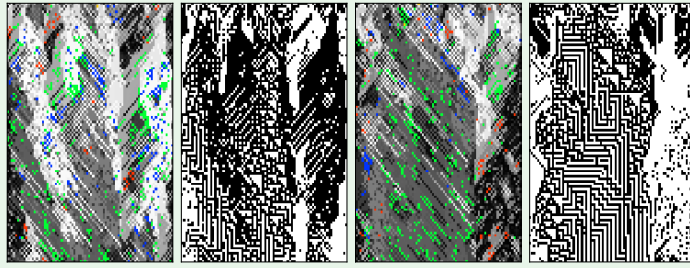
collapses ~t45



offset -3

gcd 1 · one 8-cycle

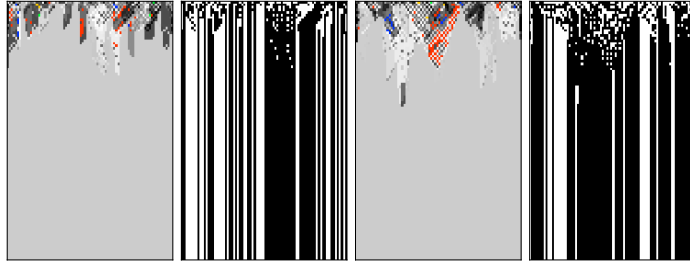
no collapse (~t100)



offset -2

gcd 2 · two 4-cycles

collapses ~t28



offset -1

gcd 1 · one 8-cycle

no collapse (~t100)

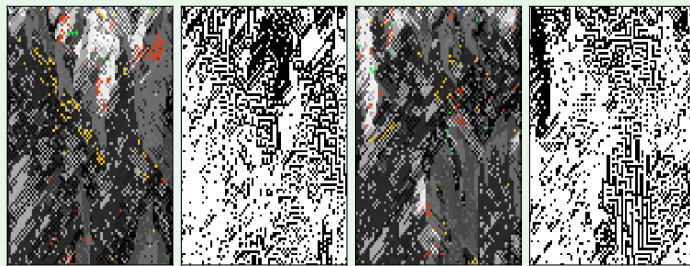


Figure S3.1

(a) *Negative offsets.* The rotation subfamily, ordered by offset: each row is one wrap rotation $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$ (two seeds; genotype in 256 colours, phenotype black and white, first hundred generations). Of these all-even rotations only the single 8-cycles, $\text{gcd}(\text{offset}, 8) = 1$ —offset ± 1 and ± 3 , highlighted—sustain a high-diversity phase; offset ± 2 (gcd 2) and ± 4 (gcd 4) collapse. See Worked example 4; continued in (b).

A family of operators, ranked by cycle structure (b · positive offsets)

Each row: $\text{bit}[\sigma(k)] := \neg \text{bit}[k]$, $\sigma = \text{rotate-by-offset (wrap)}$. genotype 256-colour | phenotype b/w, 100 gens, two seeds.

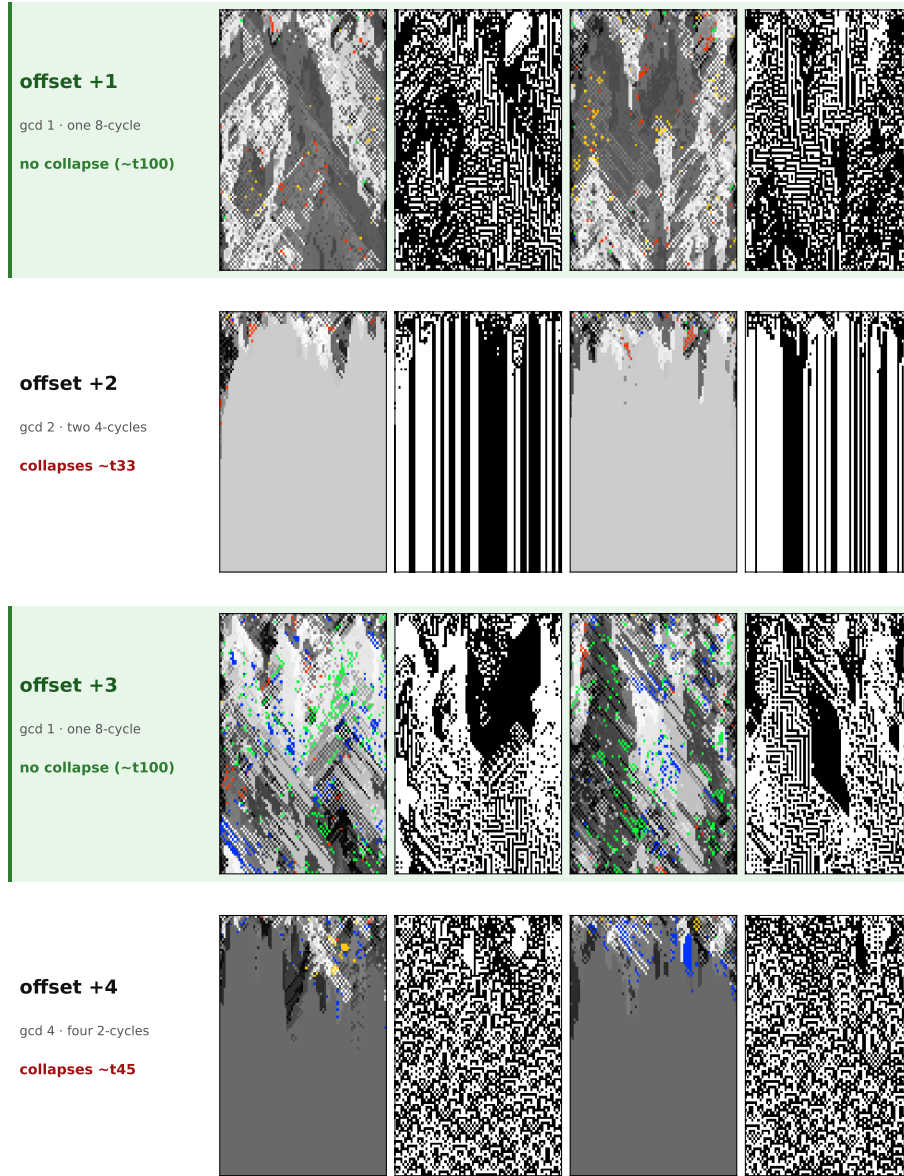


Figure S3.1

(b) Positive offsets. Continued from (a). The classification is symmetric under $d \rightarrow -d$ (verdict follows $\gcd(d, 8)$), so offset +1 sustains as offset -1 does, and offset +2 collapses as offset -2 does—inverse rotations sharing verdict but not their transients (Worked example 4). Possessing a fixed point and staying near it are different properties: the two pages visually summarise the distinction between fixed-point existence and spatial persistence.

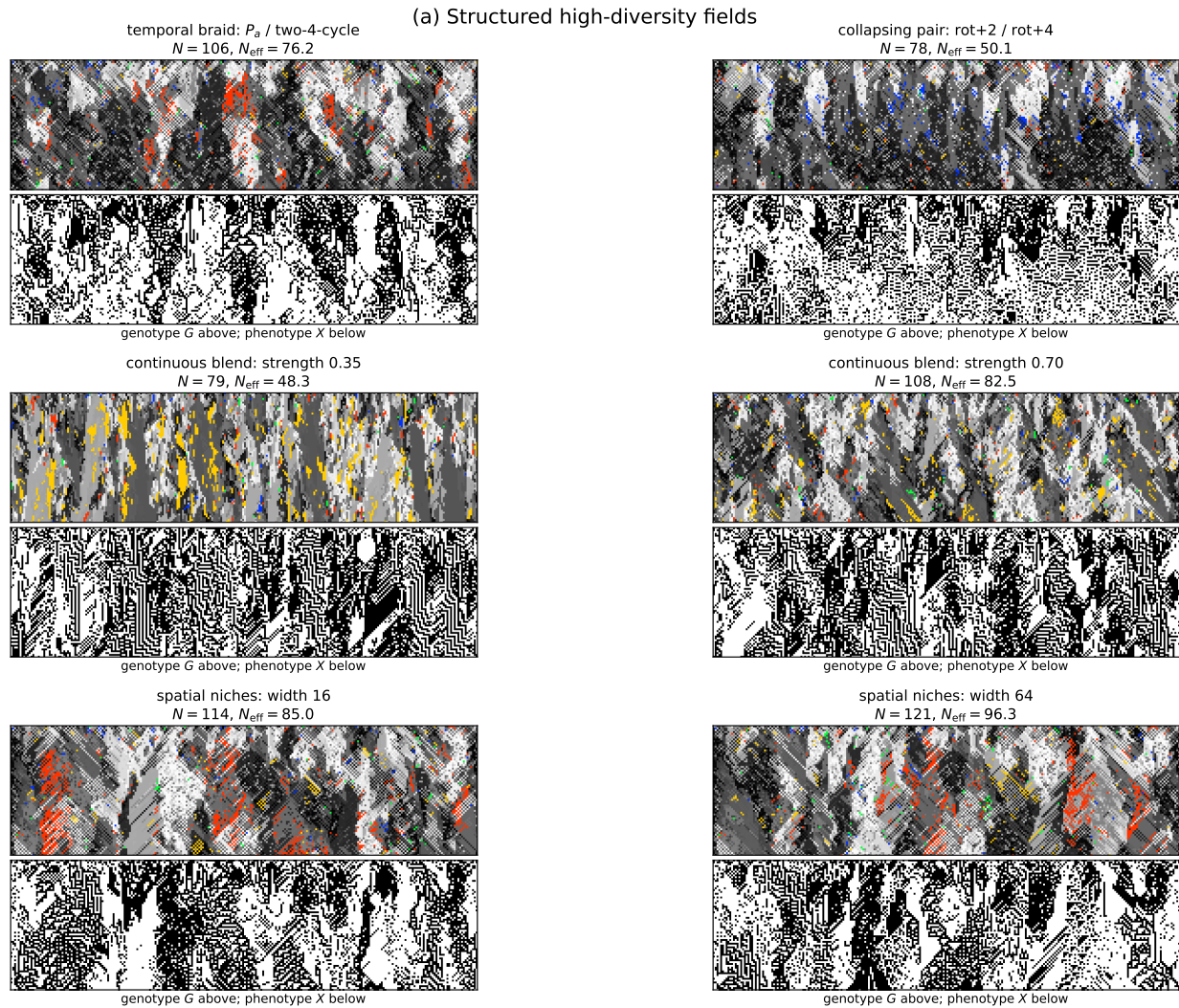


Figure S3.2

(a) *Structured high-diversity fields.* Six examples from four mechanism families: temporal alternation; alternation of two operators that collapse when run alone; continuous blending at two strengths; and spatial niches at two widths. Each tile stacks genotype over phenotype and reports both distinct-rule and Shannon-effective diversity at $T = 70$; full-population seed 0, $L = 256$. The fields sustain 78–121 rules and carry coherent diagonal structure at a range of scales. These examples motivated the candidate diversity coordinate, but the conservative-flow control in Figure 16 shows that diversity itself does not account for their propagation. Continued in (b).

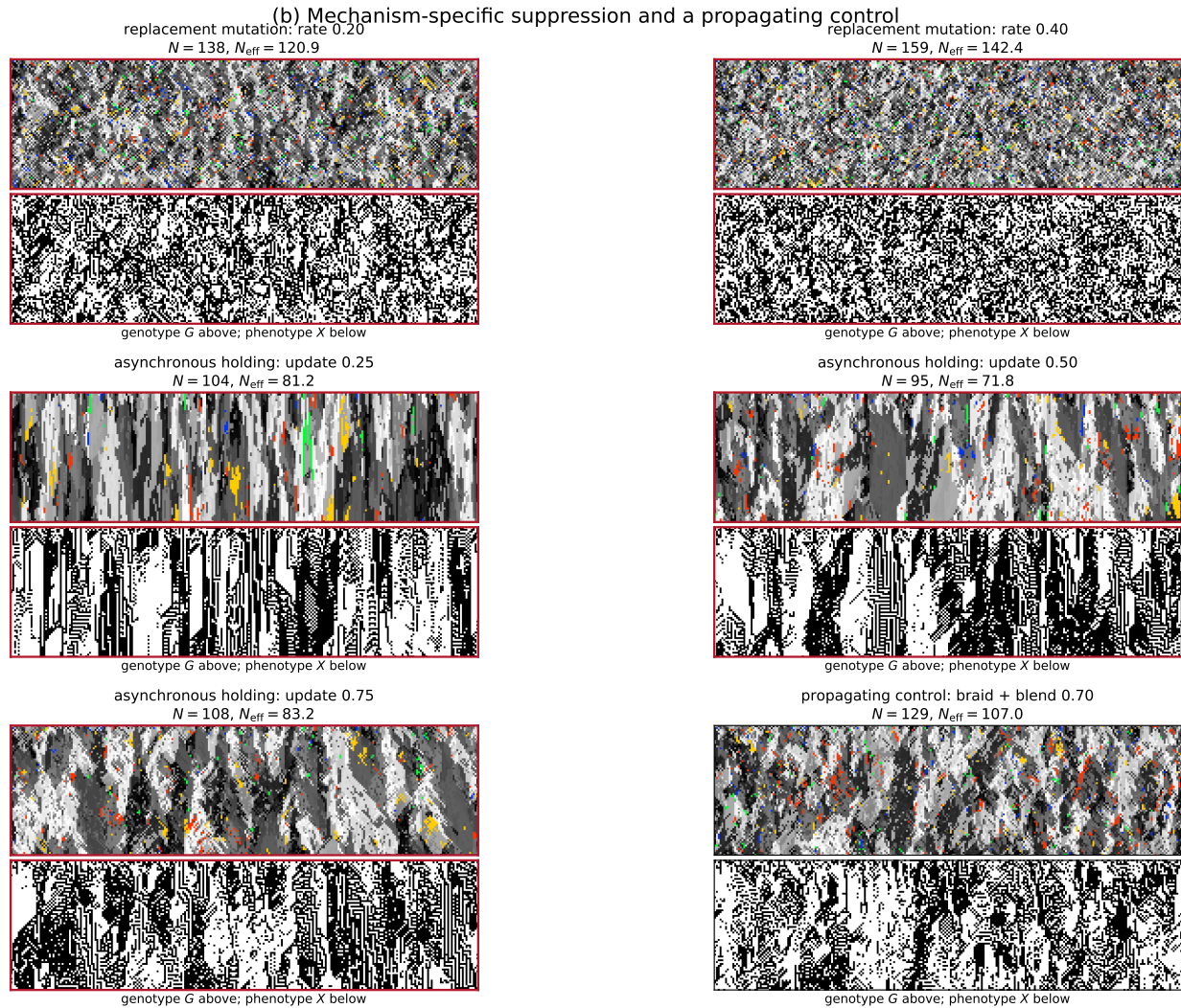


Figure S3.2

(b) *Two mechanism-specific suppressions.* Continued from (a); suppressed configurations are outlined in red. Replacement mutation at rates 0.20 and 0.40 turns the genotype into progressively finer, unstructured speckle. Asynchronous holding at update fractions 0.25, 0.50, and 0.75 instead produces vertical persistence whose strength changes continuously with the dial. The unoutlined braid-plus-blend control at bottom right propagates (3.88 cells) and recovers the diagonal texture of (a). The expanded population makes the distinction visible as a family resemblance rather than a single selected contrast: these are genuine failure modes of their respective mechanisms, not evidence for suppression by high diversity.

Supplement 4: Exploratory Domain Analysis and a Withdrawn Statistic

This second supplement carries three strands of the companion paper that are reported in full here rather than in the main text: the spatial domain structure of the genotype field and its filamentary boundaries; a measure of information transport along those boundaries that we built, validated against three nulls, and then withdrew; and the domain/particle reading that motivated both.

The main paper’s claim is about a single edge of chaos, reached by causal measurement rather than by tuning a parameter. The material here is not that. It is a real spatial structure that we spent some time mistaking for a third edge, and the record of how that mistake was made and caught. We report it at length because the failed construction is a natural one to reach for, and because the geometry-preserving nulls that appear to guard against its failure do not.

Figure, section and empirical-finding numbers below that are not preceded by “Supplement” refer to the main paper, and resolve to its numbering.

Domain Structure and Filamentary Boundaries

The finite-size scan of the main paper (*Supplement 1, Box 6*, Figure 10) finds a broad crossover rather than a critical point. We then asked whether the sustained diversity is spatially organised. It is.

Empirical finding 13: A finite-width intermediate-activity band

An intermediate-activity band ($0.15 \leq a_G \leq 0.60$) has a *lower* edge robustly at $q \approx 0.1$ —stable under both system size and threshold—while its upper edge is threshold-dependent, because a_G saturates gradually rather than jumping (the band spans $q \approx 0.1$ to 0.4 – 0.5). This finite-width crossover is what we call a *littoral edge*: a descriptive name

for a shore of finite width, not a thermodynamic phase boundary.

Definition 1 (Isolated-rule activity score). *The activity score of a rule is the mean per-step cell-change fraction over the final 150 steps of a 300-step ordinary ECA run on a periodic line of width 300. Every rule uses the same density-1/2 pseudorandom initial condition (seed 1). This finite-run measure is distinct from the static truth-table balance λ (Every Fixed Point Is Balanced). Ordered rules score near 0; Rule 110 scores approximately 0.42, and several standard chaotic rules score approximately 0.50. The score orders activity levels but does not by itself distinguish complex from chaotic dynamics.*

Tinting the genotype by this score (Figure S4.1) resolves the intermediate field into coexisting low- and high-activity domains whose boundaries trace filamentary sets through 1+1-dimensional spacetime—objects *analogous to* the domain boundaries of computational mechanics (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001), here obtained by thresholding a scalar tint rather than by predictive equivalence.

Empirical finding 14: Filamentary walls have elevated genotype churn

The activity field resolves into ordered and chaotic domains separated by *fractal-like walls* (Figure S4.2; box-counting dimension $D \approx 1.5$ – 1.7 over $L = 128$ – 768 , a handful of seeds). Those walls are the locus of elevated *genotypic* change: measuring the row-to-row rewrite rate by region (Table S4.1), the wall churn exceeds the chaotic interior in all 32 committed operator/width/seed fields and exceeds *both* interiors in 24. At $L = 256$, offset +1 gives 0.80 at the wall against 0.81 (ordered) and 0.73 (chaotic); the corresponding wall is the maximum region for $\sigma = 16250374$ and the river. The chaotic domain has the lowest aggregate churn for every operator and width (and in 29 of the 32 individual fields), so the wall—where ordered abuts chaotic—is consistently more rewritten than the chaotic interior, though not universally more than the ordered one. The walls are not compositionally special (no rule class enriched; Rule 110 is, if anything, depleted), so the distinction is dynamical rather than a difference in rule

composition. The interface analysis is exploratory: it uses a handful of seeds and a fixed-boundary variant, and the denser river boundary ($D \approx 1.8$ versus 1.5–1.7) is an unmatched comparison that does not establish an effect of genotype–phenotype coupling. Whether rewriting transports information *along* the walls is taken up in *A Measure That Failed*; the answer there is negative.

The partition is by activity, so the two sides differ in activity by construction. The question that is not self-answering is whether they differ in anything else.

Empirical finding 15: The domains hold different rules at matched activity

Restricting to cells whose *individual* activity score lies in a narrow band, and comparing the distribution over rule identities on the two sides, the total-variation distance exceeds a noise floor—obtained by randomly halving the larger group—by 13 to 25× across bands: 0.489 against 0.023 for scores near zero, 0.147 against 0.011 in $[0.20, 0.35)$, 0.295 against 0.020 in $[0.45, 0.55)$, and 0.522 against 0.021 above 0.55. Rule diversity differs as well: at matched activity in $[0.45, 0.55)$ the active domain holds 24 distinct rules per thousand cells against the quiet domain’s 38. Two cells of equal activity belong to systematically different rule populations according to which domain they occupy, so the partition captures genotypic structure the activity score does not determine. Measured on the $q = 0.25$ coexisting-domain field of Figure 10.

A Measure That Failed

If the filaments are domain boundaries they are candidate *wires*. The natural question is not which rules compose them—in our data the complex rules are not enriched there—but whether information is *transported along* them and *combined where they collide* (Mitchell et al., 1993). We built an estimator for both, and both are withdrawn. This section reports the attempt, the controls that appeared to validate it, and the control that defeated it. We keep it in the main text because the construction is one a reader might

Table S4.1

Genotype churn by region—row-to-row rewrite rate of the rule field (fraction of cells whose ECA rule changes from the row above), at $L = 256$, mean over the committed seeds. The wall exceeds the chaotic interior for every operator; for offset +1, the ordered interior is slightly higher. Wall extraction is the threshold-and-smooth procedure of Figure S4.2; reproducible from the generated fields (`plot_eoc_churn.py`).

Operator	Wall	Ordered interior	Chaotic interior
offset +1	0.80	0.81	0.73
$\sigma = 16250374$	0.71	0.68	0.57
river	0.99	0.97	0.96

reasonably build, and because the nulls that seem to guard against its failure do not.

We read both notions off the binary activity field as local transfer entropy (Lizier et al., 2008): the apparent transfer entropy from a spatial neighbour into a cell, conditioned on the cell’s own past, scores directed *transport*; the *synergy* of the two neighbours—their joint predictive contribution beyond the sum of their separate ones—scores *combining* (Figure S4.3).

A raw on-filament average will not support either claim, and we say why before using one. The score has two failure modes, and each needs its own null. First, the plug-in estimator is positively biased, so a field can score positive transfer entropy with no directed coupling at all; the null for this is a *source time-shuffle*, which preserves the marginals and the cell’s own past and destroys only the neighbour-to-target coupling. Second—and this is what defeats the raw statistic—the extracted filament is not a fixed object across a parameter sweep. It swells from 20% of the field at the bare operator to 39% under strong noise, at which point it is no longer a wall, and any on-filament mean drifts toward the field mean for reasons that have nothing to do with transport. The null for this is a *mask-matched* draw: the same number of cells, taken anywhere in the field. We report the

difference from that draw and call it *filament-specific* transport or combining; it is zero when the filament is no better than an arbitrary set of cells of the same size, and negative when it is worse.

A size-matched draw preserves the filament’s extent but not its shape, so we also ran two nulls that preserve its geometry. The first circularly translates the mask along x and averages over all displacements beyond the point where a mask ceases to overlap itself; this retains the mask’s exact connectivity, anisotropy, cell count and per-row occupancy, destroying only its registration against the field, and it involves no randomness at all. The second scores each field under the filament extracted from an independent seed at the same sweep point. Both agree with the size-matched draw to within 0.001 bits at every point ($r > 0.999$ across the sweep), and every result below is unchanged under all three. A filament-shaped region placed generically therefore scores no differently from scattered cells: what distinguishes the real filament is where it lies, not the shape of the region measured. We quote the size-matched null throughout.

The estimator has a third failure mode that no null removes, and it bounds what follows. Where the activity field is nearly degenerate—almost all cells in one state—the conditional probabilities driving the score become extreme and it inflates without bound, and this happens at *either* end. The genotypically dead offset +4 operator is the demonstration. Its genotype freezes (row-to-row rewrite rate falls to 0, Empirical *Supplement 1, Box 3*) into a population that is 98% active, and on that saturated field it records the largest on-filament transport we measured anywhere: $317\times$ the domain interior, +0.59 bits filament-specific, for an operator that transports nothing at all. The sparse end behaves the same way, and costs us the most ordered sample in the sweep below (all blend, $q = 0$, 9% active), where the shuffled surrogate outscores the real field (+0.144 against +0.105 bits, $t = -0.71$, $p = 0.49$); that point is bias-dominated and is excluded. Local transfer entropy is therefore comparable only across fields of similar and non-degenerate activity density—which is the concrete content of what earlier drafts

recorded as a “sparsity effect”.

Within these bounds the score behaves as one would hope, and it is worth recording what that looked like, because it is what made the result credible. Over the eight seeds of the $q = 1$ sweep column, on-filament transfer entropy is $+0.158 \pm 0.005$ bits against a shuffled surrogate of $+0.002$ ($t = 33.5$, $p < 10^{-8}$) and a mask-matched draw of $+0.061$, leaving $+0.097 \pm 0.005$ bits filament-specific. On the fixed-boundary field of Figure S4.2 the measure concentrates $5.7\times$ over the domain interiors; the Rule 110-dominated $\sigma = 16250374$ field, taken as a positive control, shows the same signature ($7.6\times$, filament-specific $+0.161$ bits). Three nulls agree to within 0.001 bits, the positive control behaves as a positive control should, and the effect is large and highly significant. None of that is evidence, for the reason given next.

The measure does not survive its own controls

The score defined above fails a control we did not originally run. Local transfer entropy here is computed on the thresholded activity field $\text{act} = (\text{smooth} > 0.35)$, while the filament mask is the *gradient of that same field*. The mask therefore marks precisely the cells at which act changes, and transfer entropy on a binary field is near zero except where it changes: $+0.8934$ bits on changed cells against $+0.0080$ elsewhere, the wall capturing changed cells at $163\times$ enrichment. A positive filament-specific score is guaranteed by the construction rather than by the dynamics.

Two tests establish this. Holding the mask at 0.35 and recomputing the measure at a different threshold collapses the score from $+0.0885$ to $+0.0007$ (at 0.30), -0.0044 (0.40) and -0.0377 (0.50). Deriving mask and measure from the same threshold and sweeping it instead recovers the effect at *every* threshold, growing monotonically from $+0.065$ at 0.25 to $+0.173$ at 0.50. A physical quantity should not track an arbitrary analysis parameter monotonically.

The three nulls do not defend against this and could not have. Size-matched, shift and cross-seed each control a property of the *mask*—its cell count, its shape, its

provenance—and every one of them displaces the mask off the change-cells. All three therefore report a large effect *because* of the circularity rather than despite it. This is a gap in the null design, not a fault in the nulls as implemented.

Nor is it ordinary sensitivity to how one draws a region. Four independent drawings, scored on the same fields with size-matched masks, order themselves by how much construction each shares with the measure: the thresholded gradient gives +0.0885, the gradient of the *continuous* activity field +0.0200, local causal states +0.0045, and a rule-identity boundary +0.0007. Genuine algorithm-dependence would scatter; this ranks.

Replacing the measure while keeping the mask also returns nothing. On the phenotype field, which shares no threshold with the mask, the filament-specific score is negative on all six seeds (mean -0.011) and -0.160 on the two-4-cycle field.

We therefore withdraw both claims—that these boundaries carry directed transport, and that they combine information where they meet. Combining is the synergy of the same two neighbours read off the same thresholded field under the same mask, so it inherits the circularity intact, and it likewise fails the decoupled test (phenotype-field combining is +0.003 against the size-matched null, against +0.028 for the activity field). With them goes the sweep that rested on the same statistic. What survives is reported in *A Parameter-Space Test*: the walls are where genotype rewriting is fastest, and they separate genotypically distinct populations. Those rest on different measurements—local transfer entropy is uncorrelated with genotype churn ($r \approx 0.02$)—and do not inherit this failure.

Domains and Particles

The structures reported above have an ancestor. Computational mechanics reads a cellular-automaton spacetime diagram not as one pattern but as a decomposition: *domains*, the regular backgrounds a rule settles into; *particles*, the localised structures that separate one domain from another and propagate through the diagram; and *collisions*, the interactions between particles, where the system’s computation is usually held to be taking place (Crutchfield & Hanson, 1993; Hanson & Crutchfield, 1997). Rule 110’s gliders are the

canonical case. *The Object* places this paper’s *object* among cellular-automaton variants; this section places its *structures* among ways of reading them.

The correspondence runs at all three levels. Tinting the genotype by activity resolves the intermediate field into coexisting low- and high-activity regions (Figure S4.1): domains, in the sense above, though not obtained in the same way. The filamentary walls separating them are localised, persistent, and propagate through spacetime with box-counting dimension 1.5–1.7 (Empirical finding 14); they occupy the position particles occupy, and they are where genotype rewriting is fastest. The river (Empirical *Supplement 1, Box 7*) produces coherent bands that drift through a disordered background and then branch, merge, and reorganise—which is what a population of interacting particles looks like. That last resemblance is qualitative and we do not press it: the river’s denser boundary is an unmatched comparison and establishes nothing about genotype–phenotype coupling.

Where this departs from the ancestor is the reason the walls move. In the classical setting one fixed rule runs everywhere; a domain is a pattern in the phenotype, and a particle is a defect in that pattern, carried along by the single rule that generates both. Here the domains are regions of *different rules*, and a wall is mobile because the genotype is being rewritten underneath it. Their particles move through a static rule; ours move because the rule itself is being edited at the boundary. The two pictures agree about where to look and disagree about what is doing the moving.

We inherit the vocabulary, not the machinery. Computational mechanics obtains its domains by predictive equivalence, building the ϵ -machine of the spatial configuration and reading regularity off it (Crutchfield & Young, 1989; Shalizi & Crutchfield, 2001); ours come from thresholding a scalar activity tint, which is cheaper, coarser, and carries a caveat sharper than the one usually recorded: a statistic computed on a region drawn from the same thresholded field cannot be trusted, as *A Measure That Failed* shows at length. What the tint supports is the partition itself (Empirical finding 15), not quantities evaluated on

its boundary. The mapping above is therefore a reading aid rather than a derivation. A properly computational-mechanical treatment of this substrate—domains defined by predictive equivalence over the joint genotype–phenotype configuration, particles as their boundaries—remains the natural next instrument. A first attempt, retargeting a held-out local causal-state reconstruction at these fields, agrees with the tint partition only at chance ($\kappa = 0.008$ over six seeds) and finds no transport of its own; its depth and tolerance selected at the edge of the search grid, so it is not yet decisive either way.

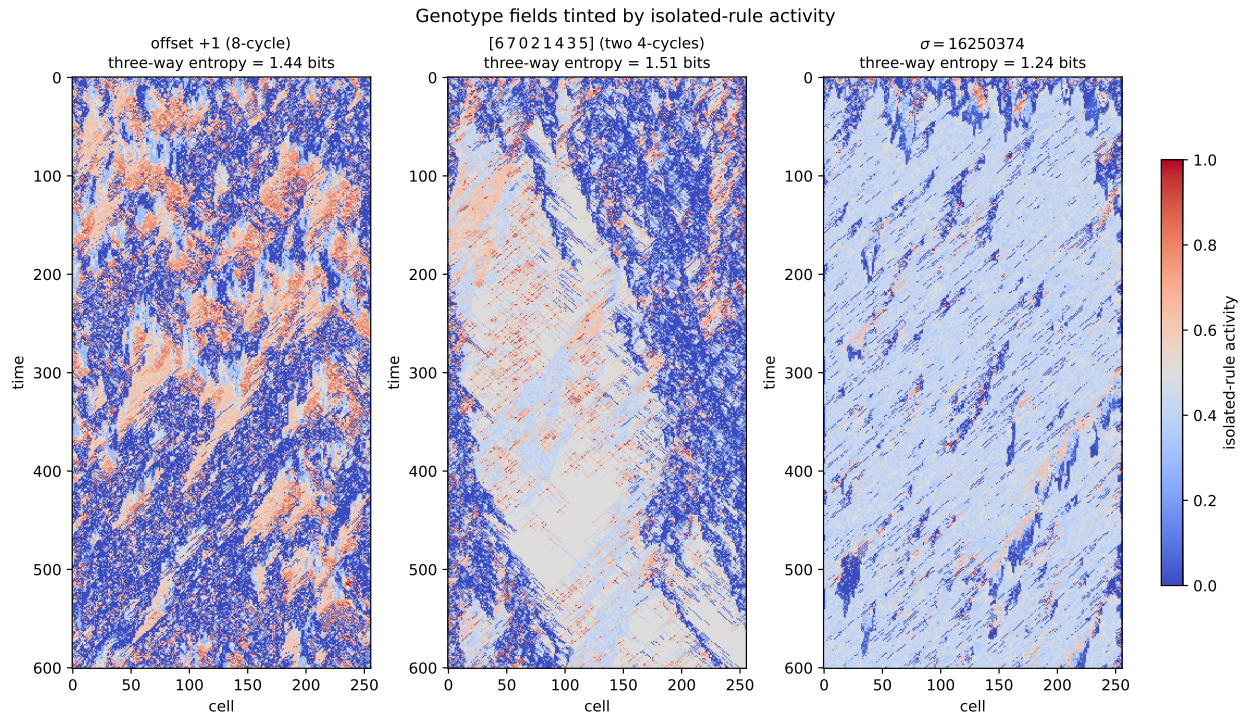
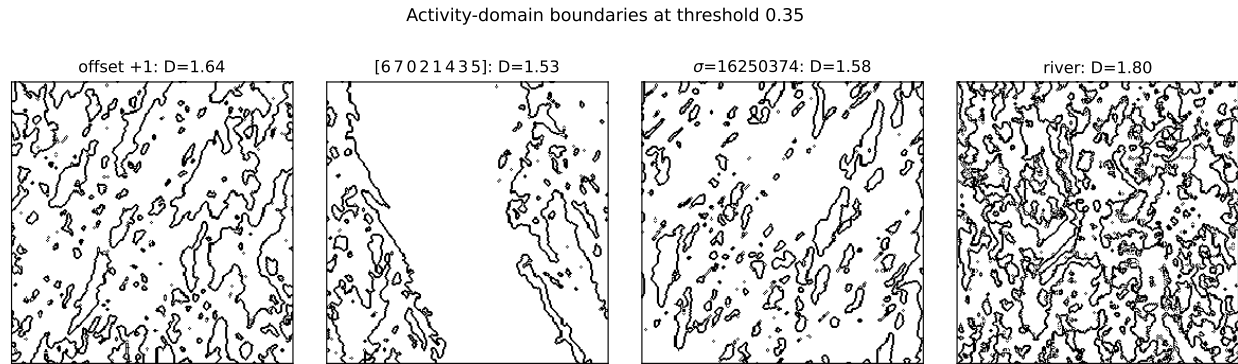


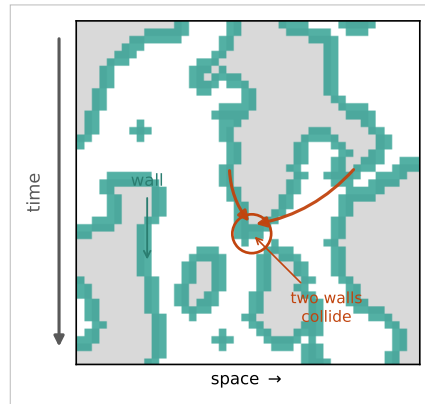
Figure S4.1

The genotype as a map of activity regimes. Genotype spacetime fields (fixed-boundary variant) tinted by each rule's isolated-rule activity score (Definition 1; blue = low near 0, pale = intermediate near 0.4, red = high near 0.5 and above). Which active regime pairs with the low-activity background depends on the operator, and the three-way balance differs sharply (entropy in bits, in title): offset +1 (the 8-cycle survivor) mixes low with high-activity chaotic rules in a fine texture; the two-4-cycle [6 7 0 2 1 4 3 5]—the standard-order form of the historical offset+2 exemplar—forms large coherent intermediate-activity domains with sharp walls, the most balanced regime mix (entropy 1.51); and $\sigma = 16250374$ is dominated by the intermediate complex Rule 110. The score orders the regimes but does not by itself separate complex from chaotic.

**Figure S4.2**

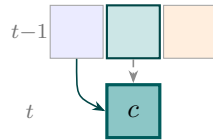
Activity-domain walls. The boundary (black) between low- and high-activity regions, traced through the 1+1-dimensional spacetime diagram at $L = 256$. Offset +1, the two-4-cycle [6 7 0 2 1 4 3 5], and $\sigma = 16250374$ carry fractal-like boundaries ($D \approx 1.5$ – 1.6); the phenotype-coupled river carries a denser one ($D \approx 1.80$, an unmatched comparison as it couples phenotype). The dimension is stable across $L = 128$ – 768 (offset +1 ≈ 1.67 , the two-4-cycle ≈ 1.50 , $\sigma \approx 1.54$, and river ≈ 1.79 ; box sizes 2–64). Collapsed operators (e.g. offset +4) carry none. These walls are also consistently more rewritten than the chaotic-domain interior (Discussion). Exploratory: a handful of seeds, fixed-boundary variant.

(a) a real patch (offset+1): activity domains and their walls



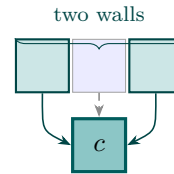
(b) transport

along a wall

*transfer entropy:*a neighbour's past
beyond c 's own

(c) combining

where walls collide

*synergy:*the two walls jointly,
beyond their
separate effects**Figure S4.3**

How the two (withdrawn) measures read a filament. The estimator illustrated here is the one this section invalidates; the diagram is retained to make the failure legible. (a) A real space–time patch from the synthetic family (offset+1): low- and high-activity domains (white/grey) meet along thin walls (teal), with one wall and one collision marked. In the schematics each cell c is predicted from its own past (dashed) and its two spatial neighbours' pasts (teal); tints mark the low- (blue) and high-activity (orange) domains a wall separates. (b) Transport is the directed transfer entropy from a neighbour into c beyond c 's own past, which the wall is defined to maximise—the circularity of A Measure That Failed. (c) Combining is the synergy of the two neighbours—their joint contribution beyond the sum of their separate ones—probed where two walls collide. History length one; conditioning on the own past removes what the wall merely carries forward in time.

References

- Anthropic. (2026a). *Claude Opus*. Retrieved July 25, 2026, from <https://www.anthropic.com/claude/opus>
- Anthropic. (2026b). *Introducing Claude Fable 5*. Retrieved July 27, 2026, from <https://platform.claude.com/docs/en/about-claude/models/introducing-claude-fable-5>
- Aracena, J. (2008). Maximum number of fixed points in regulatory Boolean networks. *Bulletin of Mathematical Biology*, 70(5), 1398–1409.
- Corneli, J., & Maclean, E. (2015). The search for computational intelligence [v1].
- Crutchfield, J. P., & Hanson, J. E. (1993). Turbulent pattern bases for cellular automata. *Physica D: Nonlinear Phenomena*, 69, 279–301.
- Crutchfield, J. P., & Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2), 105–108. <https://doi.org/10.1103/PhysRevLett.63.105>
- Denmunzio, A., Formenti, E., & Provillard, J. (2011). Non-uniform cellular automata: Classes, dynamics, and decidability [arXiv:1107.5228].
- Hanson, J. E., & Crutchfield, J. P. (1997). Computational mechanics of cellular automata: An example. *Physica D: Nonlinear Phenomena*, 103, 169–189.
- Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3), 12–37. [https://doi.org/10.1016/0167-2789\(90\)90064-V](https://doi.org/10.1016/0167-2789(90)90064-V)
- Lizier, J. T., Prokopenko, M., & Zomaya, A. Y. (2008). Local information transfer as a spatiotemporal filter for complex systems. *Physical Review E*, 77(2), 026110.
- Mitchell, M., Hraber, P. T., & Crutchfield, J. P. (1993). Revisiting the edge of chaos: Evolving cellular automata to perform computations. *Complex Systems*, 7(2), 89–130.
- Mori, T., Kudo, K., Namagawa, Y., Nakamura, R., Yamakawa, O., Suzuki, H., & Uesugi, T. (1998). Edge of chaos in rule-changing cellular automata. *Physica D: Nonlinear Phenomena*, 116, 275–282.

- OEIS Foundation Inc. (2011). Sequence A001818: Squares of double factorials; number of permutations of S_{2n} in which all cycles have even length [<https://oeis.org/A001818>; even-cycle interpretation dated 2011].
- OpenAI. (2026). *Codex*. Retrieved July 25, 2026, from <https://openai.com/codex/>
- Packard, N. H. (1988). Adaptation toward the edge of chaos. In J. A. S. Kelso, A. J. Mandell, & M. F. Shlesinger (Eds.), *Dynamic patterns in complex systems* (pp. 293–301). World Scientific.
- Pavlic, T. P., Adams, A. M., Davies, P. C. W., & Walker, S. I. (2014). Self-referencing cellular automata: A model of the evolution of information control in biological systems [arXiv:1405.4070].
- Riordan, J. (1968). *Combinatorial identities*. Wiley.
- Sakai, S., & Kanno, M. (2002). Structure of rule table and phase diagram of one dimensional cellular automata [arXiv:nlin/0204045].
- Schaller, M., & Svozil, K. (2025). Irreducible rules and equivalence classes of one-dimensional cellular automata [arXiv:2512.08117].
- Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4), 817–879. <https://doi.org/10.1023/A:1010388907793>
- Stanley, R. P. (1999). *Enumerative combinatorics, volume 2* [Problem 5.34(c)]. Cambridge University Press.
- Wuensche, A. (1999). Classifying cellular automata automatically: Finding gliders, filtering, and relating space-time patterns, attractor basins, and the Z parameter. *Complexity*, 4(3), 47–66.